A NET binding-versus-functional potency offset is a property of the assay cell, not of the assay format: consequences for external validation of monoamine transporter models

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Abstract

Background. Monoamine transporter activity models are almost universally evaluated by cross-validation. Two quantities needed to interpret one — the offset between assay modalities, and the response region an external set samples — are rarely reported, and each is a property of the data rather than of the model.

Methods. We curated 24 445 ChEMBL_37 bioactivities into a censoring-aware benchmark of 2849 compounds across 899 scaffolds and evaluated gradient-boosted models over twenty scaffold-disjoint fold realisations. Assay modality and binding radioligand were read from the free-text assay description, not from ChEMBL's `assay_type` field. The binding-versus-functional offset was decomposed by a fixed-effects model with compound-within-document effects, then stratified by radioligand. Models were tested under a protocol fixed in advance — not externally registered — against 42 stimulants from one independent study.

Results. The binding-versus-functional offset at NET is not a property of the assay format. Within compound and within document it is +0.507 in the 17 documents using [^{125}I]RTI-55 as the NET radioligand and +0.038 in the 21 using [^{3}H]nisoxetine; the pooled +0.261 is an average of the two. The same stratification at DAT and SERT spans −1.04 to +0.18, so the offset is a property of the assay cell; radioligand is collinear with chemotype here, so which feature is operative cannot be settled. The external test failed its declared criterion and was powered only for DAT. Two validation diagnostics returned numbers their construction cannot support: an outcome-conditioned error regression yields $R^2 = 0.444$ where a shrinkage null reproduces 0.448 and its only pre-outcome form gives 0.001; and a composition-matched comparator is unattainable in all nine target × set cells, though a rejection sampler returns a band for every one.

Conclusions. Three numbers here that read as properties of a model or a measurement format are properties of the set they were computed on. Report that composition — which assay cells, which response region — alongside the score, and check a matched comparator is attainable before interpreting it.

Scientific Contribution. We show that the binding-versus-functional potency offset at NET is present in assays using one radioligand and absent from assays using another, so it is not the assay-format effect it appears to be in pooled data, and that the same stratification reveals a 1.2 log-unit spread across binding assays at the other two transporters. The analysis rests on a reusable labelling of assay modality for 14 386 human monoamine transporter records derived from assay free text, now carrying measured precision against a blind annotation made under a stated rule. We further demonstrate that two validation diagnostics for external error return interpretable-looking numbers that their construction does not support, and supply the shrinkage null and the attainability check against which each should be tested.

1 Introduction

The dopamine, norepinephrine and serotonin transporters (DAT, NET, SERT; *SLC6A3/2/4*) clear monoamines from the synaptic cleft and are the molecular targets of psychostimulants, antidepressants and a continuously expanding set of novel psychoactive substances. Their relative potencies determine pharmacological character: compounds with a DAT/SERT inhibition

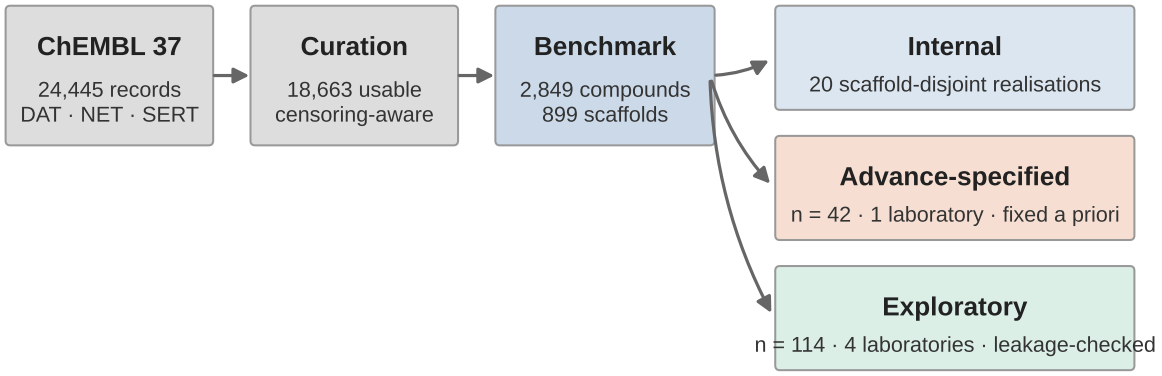


Figure 1: Study design. Flow from ChEMBL_37 retrieval through censoring-aware curation to the benchmark, and onward to its three evaluation arms. Only the second of them — the 42-compound external test — was specified before any external potency value was examined; the 114-compound extension was assembled afterwards and its endpoints were not pre-specified. Two further external sets (PDSP/BindingDB, $n = 17$, and Cui et al., $n = 20$) were assembled afterwards and are reported as an exploratory extension in § 3.12; they are not part of the advance-specified design.

ratio above 10 produce psychostimulant effects and high abuse potential, while those below 0.1 produce entactogenic, MDMA-like effects. Both numeric cut-points are stated together in the caption to Figure 1 of Luethi and Liechti's 2020 review.Luethi and Liechti (2020) The ratio itself is defined as $1/\text{DAT}\,\text{IC}_{50} : 1/\text{SERT}\,\text{IC}_{50}$ after Baumann et al. (2013), and the qualitative direction — DAT-selective compounds stimulant-like, SERT-selective compounds entactogen-like — is the older result.Luethi and Liechti (2018)

Quantitative structure–activity models for these targets are typically evaluated by cross-validation. Scaffold-based splitting, which confines all compounds sharing a Bemis–Murcko scaffold Bemis and Murcko (1996) to a single partition, is widely regarded as the conservative choice because it prevents analogue series from spanning the train/test boundary.Sheridan (2013); Lenselink et al. (2017) Recent work indicates that this expectation is optimistic. Molecules assigned different Bemis–Murcko scaffolds are frequently highly similar, so the split does not create the distribution shift it is assumed to;Guo et al. (2024, 2025) across eight datasets, fourteen model families and ten splitting strategies, scaffold-split performance was found not to differ substantially from random splitting, whereas splitting on similarity clusters was markedly harder and correlated far more weakly with in-distribution performance.Fooladi et al. (2025) Two 2026 benchmarks reach the same conclusion from different directions: that the interpolation-to-extrapolation penalty is a property of the data set rather than of the algorithm,Alshehade et al. (2026) and that splitter choice changes what an evaluation estimates far more than it changes which model wins.Crucitti

(2026) This matters for the present work in a specific way: Murcko abstraction discards ring substituents, which for this target family are the determinants of selectivity, so scaffold identity is a coarse proxy for chemical novelty here (see § 3.2 and Limitations).

Scaffold splitting addresses one leakage pathway. It does not control for assay heterogeneity, laboratory provenance, or the narrowing of response range that occurs when a benchmark is assembled from historical literature.Landrum et al. (2023); Mayr et al. (2018) Whether scaffold-split estimates predict performance on compounds measured elsewhere has not been established for this target family, and the question is practically consequential: forensic and regulatory users apply such models to newly emerging substances characterised in laboratories other than those contributing the training data.

We test this directly. The model and evaluation endpoints were fixed before any external value was examined, and the advance-specified result is reported unchanged regardless of outcome. We then decompose the observed shortfall into contributions from response range, chemotype, assay modality and test-source identity.

2 Methods

2.1 Data source and curation

Bioactivities were retrieved from the ChEMBL REST API (release ChEMBL_37, released 2026-05-01; accessed 2026-09-08) for the human single-protein targets CHEMBL238 (DAT), CHEMBL228 (SERT) and CHEMBL222 (NET), restricted to K_i and IC_{50} endpoints, returning 24 445 records.Zdrazil et al. (2024); Davies et al. (2015)

Records were removed if ChEMBL flagged a data-validity problem, if the assay type was neither binding (B) nor functional (F), if a canonical SMILES was absent, or if units were not nanomolar, leaving 18 663 usable measurements. By ChEMBL's `assay_type` field 97.6% of these derive from radioligand binding assays and 2.4% from functional assays; § 2.2 shows that field to be unreliable here and recovers a very different proportion from the assay text. Structures were standardised with RDKit RDKit (2026) and potencies converted to the p-scale ($-\log_{10}$ molar).

The resulting endpoint is heterogeneous. It combines radioligand binding affinity with functional uptake-inhibition potency, and pools transporter substrates with pure reuptake blockers. The

models therefore predict a ChEMBL-derived transporter activity endpoint rather than a uniform mechanistic quantity, and § 3.5 quantifies one consequence of that heterogeneity.

2.2 Labelling assay modality from the assay description

ChEMBL's `assay_type` field does not separate radioligand binding from functional uptake inhibition in this corpus: 8420 of the 8801 IC_{50} records in the re-retrieval described below are tagged `B`. Taken at face value the field implies that functional measurements are 2.4% of the data, on either corpus, and therefore too scarce to model, a conclusion that is an artefact of the annotation rather than of the literature. It is more accurate to say that the field under-recovers functional measurements than that it mislabels them: of the 403 records ChEMBL tags `F` in this corpus, 337 are functional uptake by the text, so the tag is specific (84%) but recovers 6% of the 5601 functional records the text identifies. These 403 are counted on the 16 750-record re-retrieval and are not the same quantity as the 2.4% functional share of the 18 663-record frozen set reported in § 2.1 (roughly 448 records); the two happen to land on similar fractions of different corpora. `assay_type` encodes how the source publication classified the assay, which is a different question from what was measured, and the gap below should be read as a definitional mismatch rather than as curation error.

Which corpus. The labelling is computed on the independent re-retrieval of § 2.11 (16 750 records, 2026-09-27), not on the frozen ChEMBL_37 benchmark of § 2.1 (18 663 usable measurements). The two counts are different quantities and were previously reported without saying so. Every offset analysis in § 3.5–§ 3.7 is therefore computed on the re-retrieval; the benchmark and all model fitting use the frozen set. The deposited labelling table carries a provenance column.

Modality was therefore read from the free-text `assay_description` by ordered, high-precision regular expressions, independently of `standard_type`. The phrases *uptake site* and *reuptake site* are neutralised first, because there the word names the target rather than the measurement; descriptions asserting both modalities are placed in a `conflict` class and undecidable ones in an `unclear` class, and both are excluded rather than guessed. Of 16 750 records, 8785 classify as binding, 5601 as functional uptake, 810 as conflicting and 1554 as undecidable — so functional measurements are 39% of the classifiable corpus, not 2.4%. Per-target compound counts with functional data are 1216 (DAT), 1732 (NET) and 1774 (SERT).

This labelling is what makes the decomposition in § 2.3 possible, and it is deposited as a reusable table of 11 693 (compound, target, modality) medians. Automated and manually curated re-annotation of ChEMBL assay descriptions has recently been pursued at corpus scale;Smit et al. (2026) the labelling here is narrower — one binary distinction over one target family. Its accuracy is measured rather than assumed. A stratified random sample of 186 distinct assay descriptions was annotated from the text alone, blind to the classifier's output, under the rule that a description counts as binding only if the measurement is displacement of a named radioligand and as functional only if it is inhibition of substrate transport, with everything else recorded as undetermined. On the 130 sampled records the classifier actually emits — it excludes the `conflict` and `unclear` classes — precision is 0.914 (95% CI 0.825–0.960) for binding and 0.967 (0.886–0.991) for functional uptake, and overall agreement is 0.938 (0.883–0.968). Recall is 1.000 for both classes, but that figure is close to tautological given the error structure and should not be read as perfect recall: it is computed only over the records the annotation resolved. The informative property is the negative one. There is no case in the sample where the classifier assigns one modality and the annotation reads the other; every disagreement is the classifier committing where the annotation judged the text undetermined, and four of the eight arise from assay text that describes a different target. The excluded classes behave as intended in one direction and conservatively in the other: `unclear` is 36 of 40 genuinely undetermined, while `conflict` is 11 of 16 recoverable functional records, so the exclusion discards real functional data rather than admitting wrong labels. The annotation was produced by a large language model under the stated rule and is deposited in full; it is a reproducible-rule annotation, not an independent expert gold standard, and is reported as such. Within-pair designs of this kind have been shown to be the robust way to work across inter-assay variability.Nelen et al. (2025) The consequence of not making the distinction has been stated for this target family: Lee et al. conclude that it is "not advisable to combine either K_i and IC_{50}, or uptake and binding data sets" when modelling the dopamine transporter, while noting that the direction of any such offset was undetermined.Lee et al. (2024)

2.3 Separating assay format from value type

Because ChEMBL's binding records are predominantly K_i and its functional records predominantly IC_{50}, a comparison of "binding K_i" with "functional IC_{50}" moves two things at once:

the assay format and the value type, the latter of which carries a Cheng–Prusoff relationship.Kalliokoski et al. (2013) With modality labelled independently of `standard_type` the two become separable, and the off-diagonal cells are well populated (3018 binding IC_{50} and 1336 functional K_i records).

Both effects were estimated in one model: pChEMBL regressed on an indicator for functional modality and an indicator for K_i, with **compound-within-document fixed effects** — so every contrast is between assays run on the same compound in the same publication — and standard errors clustered on document, the unit that shares a protocol, cell line and operator. Robustness was assessed by leave-one-document-out, by moving the `conflict` class into each modality in turn, and by refitting with document rather than compound-within-document effects.

Mechanism stratification. To test whether the format effect is carried by transporter substrates, for which $K_m \neq K_D$ is expected,Schömig et al. (1988); Eshleman et al. (1999) compounds were classified as substrate-type releasers or uptake blockers *from structure alone*. Using the K_i/IC_{50} ratio for this purpose would be circular, since that ratio is the quantity under test. Blocker-defining motifs (α-pyrrolidinophenone, tropane, aryloxy and diaryl ether, piperazine, fused tricyclic, benzhydryl, α- and β-ester, *N*-*tert*-alkyl) are applied first as hard negatives, then a molecular-weight and ring-count gate, then substrate cores (arylethylamine, cathinone, constrained amine, arylmorpholine, aminooxazoline, aminoindane). Against a 55-compound reference set — 28 substrate-type releasers and 27 uptake blockers, deposited in full with class labels — the rule achieves 96% recall on releasers, 100% specificity on blockers and 98% balanced accuracy. The single miss is MDAI, a ring-constrained releaser caught by the ring-count gate. A missed releaser contaminates the blocker stratum anti-conservatively, so the blocker-only estimate of § 3.7 is a lower bound on how clean that stratum is.

2.4 Identifying the binding radioligand

The same assay descriptions that carry the modality also name the radioligand. A second set of ordered regular expressions extracts it, matching the specific ligands first — nisoxetine, desipramine, RTI-55 (with its synonym β-CIT), mazindol, WIN 35,428 (with CFT), citalopram, paroxetine, imipramine, cocaine and GBR 12935 — before the native substrates norepinephrine, serotonin and dopamine, and recording `unstated` otherwise. The radioligand is recovered from

77–84% of binding records at each target. A document is assigned to the stratum of the radioligand it uses for the most records, and a stratum is reported only if it holds at least five documents — which excludes one cell, [^{3}H]mazindol at DAT (3 documents, 37 records, −0.036), whose inclusion would not change any statement made here.

This was added during revision, in response to the observation that a NET-specific discrepancy between binding and functional measurement has an obvious candidate cause that is not the assay format: human NET binding is dominated by a small number of radioligands whose nonspecific binding and whose K_D in the Cheng–Prusoff conversion differ, and a systematically mis-calibrated K_D in one of them produces a constant shift in every K_i from that assay, at one target only. It is a text query on data already parsed, so the hypothesis is testable without new experiments.

2.5 Censoring

Left-censored measurements were retained at their bounds. Exclusion is not neutral: discarding censored records removes 672 compounds (23.6%) non-randomly, with bounds clustering at the 10 µM screening concentration and the discarded population differing from the retained one at every target (Kolmogorov–Smirnov $p \leq 6 \times 10^{-95}$). Exclusion removes 189 scaffolds entirely and shifts the selectivity class distribution (χ^2, $p = 1.6 \times 10^{-7}$), depleting the psychostimulant-like class from 16.2% to 7.9%. Censored records are 18.2%, 7.3% and 10.5% of the DAT, NET and SERT columns, with a median bound of $p = 5.00$ against uncensored medians of 6.78, 6.89 and 7.13 — they sit at the weak end, as the screening concentration implies.

Retaining a bound as a point value is a benchmark-preserving convention, not an assertion that the latent potency equals the bound. To bound what the convention costs, the same pipeline was refitted with a censored objective that penalises predictions above the bound and leaves predictions below it free ($L = \max(0, f - y)^2$ for censored records, $L = (f - y)^2$ otherwise), over ten scaffold-disjoint realisations, scoring on uncensored compounds only so that every label is a real value. Internal r^2 changes by +0.007 (DAT), +0.003 (NET) and −0.001 (SERT) and RMSE by 0.000, +0.001 and +0.002; the full table is deposited. The choice of loss is therefore not what any conclusion in this paper rests on. Methods that treat censoring as more than a loss-function choice exist and are not pursued here: Svensson et al. show that carrying censored labels through the likelihood improves calibrated uncertainty rather than point accuracy, which

Table 1: Measurement reproducibility by target. Standard deviation and standard error of the mean across replicate measurements of the same compound at the same target, in log units.

Target	Groups	Median SD	Mean SD	Median SEM
DAT	823	0.166	0.268	0.100
NET	802	0.212	0.293	0.132
SERT	1032	0.194	0.296	0.137
All	2657	**0.194**	0.287	**0.127**

is consistent with the deltas above and is the direction in which this benchmark would benefit from them.Svensson et al. (2025) (These r^2 values, 0.523/0.465/0.502 under squared loss, are not comparable with Table 2: they are computed on the uncensored subset alone, which excludes the censored compounds clustered at the bound and is a harder set.) **Final benchmark: 2849 compounds across 899 scaffolds.**

2.6 Measurement reproducibility

A replicate group is all records for one (compound, target) pair, taken across documents, assay types and modalities, after the curation filters of § 2.1; the median group holds 2 records and the 90th percentile holds 5. Table 1 reports the median and mean of the per-group standard deviations, which describe the distribution of agreement between replicates. Where an error *variance* is needed rather than a description — in the ceilings of Table 15 — we use the pooled, degrees-of-freedom-weighted estimate instead, which is 0.444, 0.471 and 0.525 on these same groups. It is larger than the median because a two-record group's standard deviation is a low-biased estimate of the dispersion it is drawn from, and the pooled figure is the one that estimates that dispersion. Because the groups span modalities, the figures above are an upper bound on pure measurement error: they already contain part of the assay-modality offset of § 3.5, which is why that offset is compared against this floor rather than estimated from it. Restricting groups to a single modality is not possible at useful sample size, since most compounds are measured one way only.

These values sit close to the canonical estimate for heterogeneous public K_i data: Kramer et al. report a mean error of 0.44 and a standard deviation of 0.54 pK_i units across ChEMBL, after filtering for the unit-transcription errors, undifferentiated stereoisomers and repeated citations that inflate naive estimates, and derive a maximum attainable R^2 of 0.81 on large data sets.Kramer et al. (2012) Our 0.194 median SD is smaller because it is computed within one

target family and after the same class of filtering, and our 0.820/0.633/0.838 internal ceilings are the same calculation applied per target.

Because models predict the aggregated (median) value, the relevant comparator for model error is the standard error of that aggregate, 0.127 log units.

2.7 Descriptors, models and cross-validation

Features were ECFP4 Morgan fingerprints Rogers and Hahn (2010) (radius 2, 2048 bits) with ten physicochemical descriptors. Models were LightGBM Ke et al. (2017) gradient-boosted regressors (250 estimators, learning rate 0.06, 31 leaves, minimum 10 samples per leaf, column subsample 0.4, fixed seed), with hyperparameters fixed before external evaluation. They were not tuned: they are LightGBM defaults except for the learning rate, leaf count and column subsample, which were set once by inspection of scaffold-CV performance on an early version of the benchmark and never revisited. No nested cross-validation was run. That early inspection used the same kind of folds as the reported CV, so the internal estimate is a selected rather than a clean one, and it is the comparator against which the external shortfall is measured. We do not attempt to bound the optimism this introduces, because fold-seed variation at fixed hyperparameters and variation over the learning-rate, leaf-count and column-subsample grid are different spaces and the first does not bound the second. The claim is the weaker and sufficient one: a selection over three parameters, made once on an earlier version of the benchmark, cannot plausibly account for shortfalls of 0.13–0.49. The optimism is not zero, and a shortfall of 0.02 would not survive it; these are not that. Five scaffold-disjoint folds were generated with zero cross-fold scaffold overlap verified, repeated twenty times with different scaffold orderings; internal figures are mean ± SD across those realisations. Note that fingerprint features encode ring substitution, so substituent information is available to the model even though it is discarded by the Murcko abstraction used to define the folds.

Which R^2. Two quantities share the symbol and they are not interchangeable. Unless stated otherwise, values reported here as R^2 or r^2 are the *squared Pearson correlation* between observed and predicted values, which is the definition used throughout the analysis code. It is invariant to any linear rescaling of the predictions and therefore measures agreement in ranking and spread only. The coefficient of determination, $1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$, additionally penalises predictions that

are systematically shifted or compressed, and is reported separately where the two diverge — which on the external sets they do substantially, because the predictions compress toward the training mean (§ 3.4). Both are tabulated for every external cell in the deposit.

Reproducibility of the primary analysis. The internal pipeline was re-implemented from the deposited source and re-run over twenty scaffold-disjoint realisations. It reproduces Table 2 to within 0.003 in r^2 and 0.003 in RMSE at every target (r^2 0.584 / 0.502 / 0.538 against 0.584 / 0.501 / 0.541; RMSE 0.736 / 0.811 / 0.887 against 0.735 / 0.811 / 0.884). The residual difference is the fold-seed stream, not the model. Table 2 and every external quantity in this paper are reported from the *original* run; the re-run is a check on it and is not mixed into any reported figure.

2.8 External evaluation under an advance-specified protocol

The protocol below was written down before any external potency value was examined, and the result is reported unchanged. It was *not* registered with any external registry, carries no third-party timestamp and no deposited hash, so the claim is self-attested and a reader cannot verify it independently; we use "advance-specified" rather than "pre-registered" throughout to mark that distinction. Two amendments were made after the external values were seen; both are declared inline below, in the places they apply. *Endpoints, ranked in advance:* (1) DAT/SERT log-ratio regression; (2) selectivity classification; (3) per-target regression; (4) exploratory cross-target error correlation. *Exclusions:* as specified, compounds sharing a full InChIKey with any training compound were removed. *Deviation, declared:* during revision this was tightened to the InChIKey *skeleton block*, which is the correct unit for this feature set because ECFP4 ignores stereochemistry — a stereoisomer of a training compound carries a different full key and an identical fingerprint. On the advance-specified test the two rules are equivalent: both remove the same four compounds, 2Me-α-PVP, 3Me-α-PVP, cocaine and MDMA, so no advance-specified result turns on the change. It is not equivalent on one of the exploratory sets added later, where it removes 13 further compounds and lowers a reported R^2; that case is set out in § 3.12. *Success criteria:* external R^2 within 0.10 of internal → generalises; more than 0.20 below → generalisation failure, reported as a finding. *Test set:* 46 compounds from Stalberga et al.Stålberga et al. (2026), of which 4 were excluded by either rule, leaving 42. *Amendment,*

declared: one of the 46 structures was found during revision to be the wrong molecule — cocaine recorded as its inactive hydrolysis metabolite benzoylecgonine — and was corrected. Because the training set holds (−)-cocaine, the corrected structure is caught by the overlap filter and the wrong one was not, so the correction moved the test set from 43 compounds to 42. Both the error and its two consequences are set out above. A cross-check of all 40 external structures carrying PubChem identifiers found no further discrepancy beyond free-base-versus-salt differences, which are intended. We note what this means for the status of the protocol: a specification that can be amended after the outcome is known, with no external timestamp to anchor the original, is weaker evidence than a registered one, and we claim no more for it than that.

Exploratory extension. After the advance-specified analysis was complete, the external set was extended with the Luethi and Liechti compilation,Luethi and Liechti (2020) six Basel primary studies,Simmler et al. (2013, 2014); Rickli et al. (2015a,b); Luethi et al. (2019); Kolaczynska et al. (2019) and Eshleman et al.Eshleman et al. (2017) Structures were resolved by two independent routes: CAS registry lookup and OPSIN parsing of IUPAC names.Lowe et al. (2011) Analyses in § 3.9 and § 3.10 are exploratory.

2.9 Use of large language models

Analyses were designed, implemented and executed with assistance from a large language model (Claude, Anthropic) acting as a coding and analysis collaborator. The model did not author the manuscript and does not meet authorship criteria. All numerical results reported here were produced by executed code and are reproducible from the deposited repository. Implementation errors identified and corrected during development are enumerated in the Declarations.

2.10 Decomposition of external error

Absolute prediction error was regressed on three predictor blocks: response-range position (distance beyond the training interquartile range, zero inside it), chemical similarity (nearest-neighbour Tanimoto to training), and source identity (dummy-coded by study). Each block's *unique* contribution is the drop in model R^2 when that block alone is removed.

2.11 Independent re-retrieval and chemotype stratification

To test whether the assay-modality offset is an artefact of a single retrieval or of a single chemotype, activities were retrieved again from the live ChEMBL REST API on 2026-09-27, sixteen months after the frozen ChEMBL_37 release, using an independently written curation script. Binding was taken as K_i from assays ChEMBL types `B`; functional as IC_{50} from assays typed `F`; records were restricted to human targets with a ChEMBL `pchembl_value` present, an uncensored relation, and no data-validity comment. Per compound and modality the median was taken, and the offset defined as functional minus binding. This retrieval returned 16 750 records and yields a compound set that is *not* nested within the frozen one; it is a robustness check on independent data, not a reproduction of the frozen estimate.

The resulting NET paired set was stratified two independent ways: Butina clustering Butina (1999) of ECFP4 fingerprints at 0.6 Tanimoto, and grouping by Bemis–Murcko scaffold. Strata with fewer than five compounds are not reported. Confidence intervals are percentile bootstrap over compounds (4000 resamples). A leave-one-stratum-out analysis recomputes the pooled offset with each stratum deleted in turn.

2.12 Two competing explanations for the NET offset

A difference between binding and functional values for the same compound need not arise from assay modality. Two alternatives were tested.

Laboratory of origin. If the binding and the functional values are contributed by systematically different papers, the offset could be a between-study calibration difference. Each activity record carries its source document, so the analysis was repeated on the subset of compounds for which *both* modalities come from the same ChEMBL document — the same laboratory, the same compound series, the same report — which removes between-study variation by construction. Pooling potency values across sources without such a restriction is itself a documented source of noise.Landrum and Riniker (2024) A leave-one-document-out analysis recomputes the pooled offset with each contributing document deleted in turn.

Substrate-type releasers. The ratio of binding affinity to uptake potency is itself the field's standard discriminator between substrate-type releasers and pure uptake blockers,Eshleman

et al. (2017) and releasers show ratios one to two orders of magnitude above blockers at NET. The offset under test *is* that ratio, so labelling compounds by the ratio would assume the confound rather than test it. Three independent routes were used instead: (i) presence of any release or efflux assay record for the compound in ChEMBL, identified from assay text; (ii) the substrate-type releaser pharmacophore applied to structure alone — an acyclic aryl–C–C–N chain with a primary or small secondary amine, which excludes tropanes, methylphenidate analogues, aminotetralins and the α-pyrrolidinophenones; and (iii) identity against literature-established releasers and blockers.

3 Results

3.1 Internal performance

DAT/SERT log-ratio regression gave $R^2 = 0.564$. Classification reached AUC 0.833 ± 0.013 (psychostimulant-like) and 0.847 ± 0.005 (entactogen-like), against permutation nulls at 0.495 ± 0.021 and 0.501 ± 0.018 — 16–19 standard deviations above chance. Internal prediction errors exceeded the within-compound replicate SEM several-fold. That SEM does not capture between-assay or between-laboratory variability: cross-assay dispersion measured below (§ 3.5, SD 0.497–0.658) is three to five times larger. Within-compound repeatability is a lower bound on measurement variability, not a ceiling on achievable accuracy.

3.2 Scaffold splitting reduces but does not eliminate leakage

Scaffold splitting is adopted on the premise that it separates analogue series, and it does so only partly here. Under random splitting 74.5% of test compounds have a training compound within 0.7 Tanimoto; under Bemis–Murcko scaffold splitting 26.2% still do (Figure 2b). The reduction is real and the residue is not small: better than one test compound in four is a near neighbour

Table 2: Internal cross-validated performance. Mean ± SD across twenty scaffold-disjoint fold realisations.

Target	R^2	RMSE	95% range (R^2)
pDAT	0.584 ± 0.013	0.735 ± 0.012	$[0.570, 0.610]$
pNET	0.501 ± 0.013	0.811 ± 0.011	$[0.476, 0.519]$
pSERT	0.541 ± 0.013	0.884 ± 0.012	$[0.518, 0.564]$

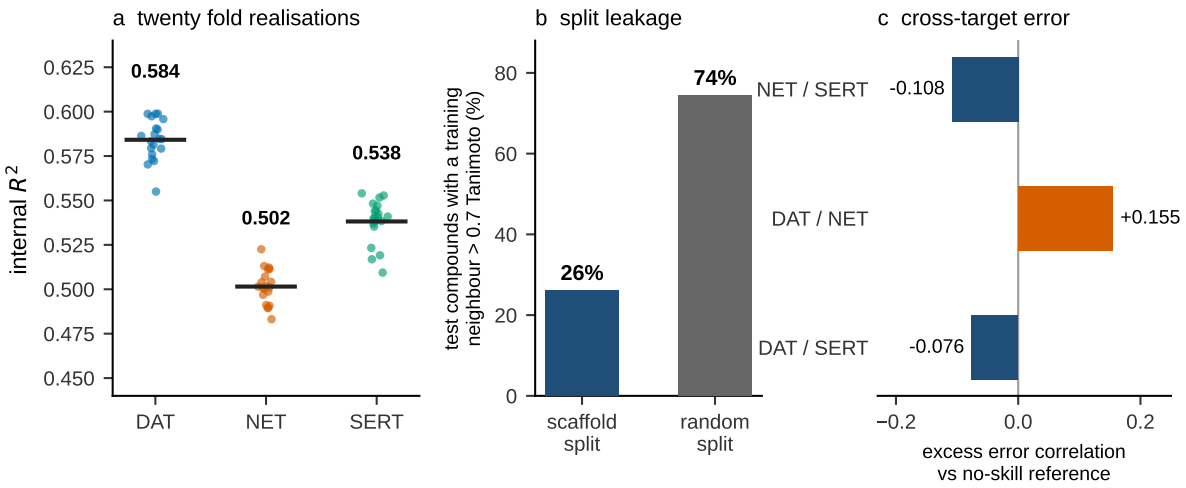


Figure 2: Internal performance and split behaviour. (a) R^2 across twenty independent scaffold-disjoint fold realisations; horizontal bars are means. Panel (a) is drawn from the independent re-implementation of § 2.7, because that is the run for which the per-realisation values were retained; its means are 0.584 / 0.502 / 0.538 against Table 2's 0.584 / 0.501 / 0.541, which are the originals and are what every reported figure uses. (b) Percentage of test compounds having a training neighbour above 0.7 Tanimoto under scaffold versus random splitting: scaffold splitting reduces near-neighbour leakage but does not eliminate it. (c) Excess error correlation between target pairs against a no-skill reference; positive values indicate shared model bias. Error bars are SD across realisations.

Table 3: Advance-specified external evaluation. Bootstrap 95% confidence intervals from 2000 resamples. n.s. = not significant.

Endpoint	Internal	External ($n = 42$)	95% CI	p
DAT/SERT ratio (primary)	0.564	0.114	—	0.029
pDAT	0.584	0.459	$[0.209, 0.690]$	< 0.0001
pNET	0.501	0.015	$[0.000, 0.150]$	0.44 (n.s.)
pSERT	0.541	0.058	$[0.000, 0.284]$	0.12 (n.s.)

of something the model was fitted on, which is the quantity scaffold splitting is usually taken to have removed. The reason is the one given in the Introduction: Murcko abstraction discards ring substituents, and for this target family those substituents carry much of the selectivity, so two compounds with different scaffold strings can be close analogues.

Errors are also correlated across targets beyond what independent models would produce. Measured against a no-skill reference, the excess error correlation is +0.155 for DAT/NET, −0.076 for DAT/SERT and −0.108 for NET/SERT (Figure 2c). The positive DAT/NET value indicates shared bias rather than shared signal, and that is the same pairing which carries the assay-modality offset of § 3.5.

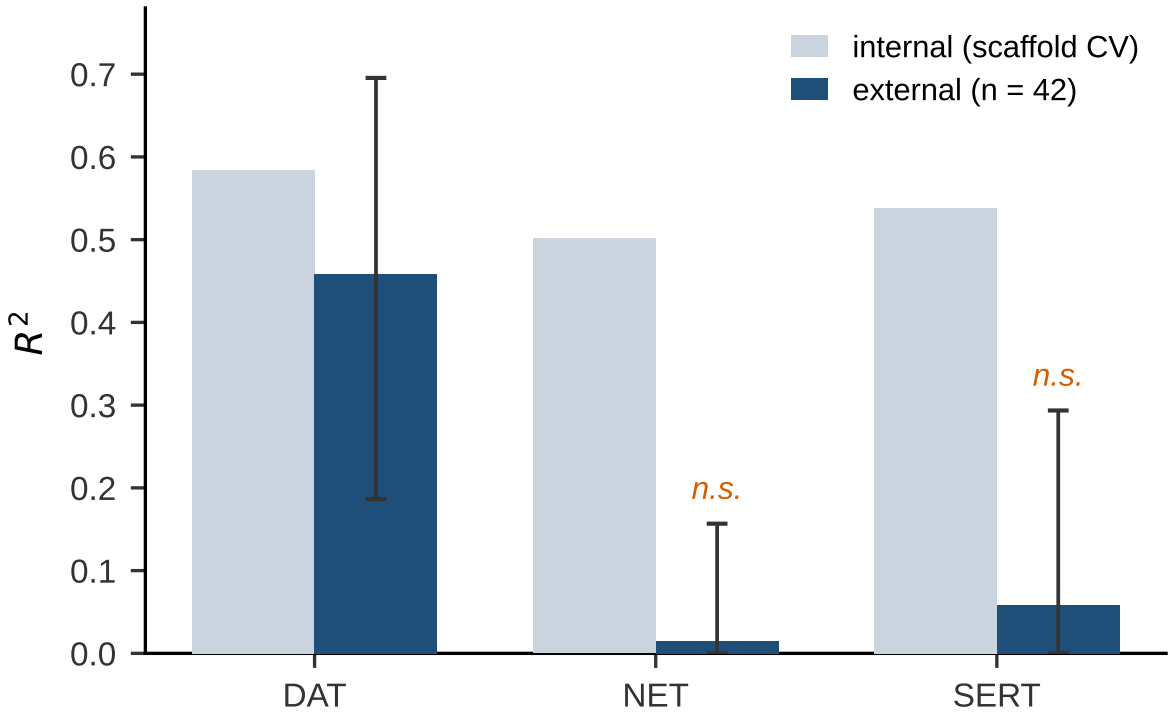


Figure 3: Advance-specified external test. Internal versus external R^2 with bootstrap 95% confidence intervals; *n.s.* marks correlations not distinguishable from zero. The companion panel of post hoc sample-size calculations has been removed: observed-power figures are a monotone transform of the p-value and add nothing the intervals do not already carry.

3.3 The advance-specified external evaluation

The advance-specified primary endpoint failed its declared criterion, falling more than 0.20 below the internal estimate. At $n = 42$, only DAT was adequately powered: detecting correlations of the observed magnitude would require $n = 15$ for DAT, $n = 133$ for SERT and $n = 515$ for NET. These are post hoc calculations based on observed effects and largely restate the confidence intervals; the defensible statement is that the available sample could not distinguish null from modest external association at NET and SERT. Leakage does not explain the shortfall — median nearest-neighbour Tanimoto to training was 0.505, and no external compound exceeded 0.85.

3.4 Decomposition of the shortfall

Range-restricted retraining — a sensitivity analysis rather than a decomposition, since it selects training data using the external response distribution — reproduces 90%, 53% and 60% of the shortfall at DAT, NET and SERT (bootstrap 95% CI for NET $[51, 73]$); chemotype matching

Table 4: Matched-training decomposition. Internal cross-validated R^2 under training sets matched to the external set on response range, chemotype, or both. The Baseline column is a *single* scaffold-CV realisation — the one this experiment was run on — not the twenty-realisation mean of Table 2 (0.584 / 0.501 / 0.541). The difference is within the realisation-to-realisation spread of ±0.013 reported there, and the comparison within this table is between columns computed on the same realisation, which is what the decomposition requires.

Target	Baseline	Range-matched	Chemotype-matched	Both	External
pDAT	0.593	0.472	0.576	0.450	0.459
pNET	0.511	0.248	0.458	0.195	0.015
pSERT	0.552	0.257	0.474	0.249	0.058

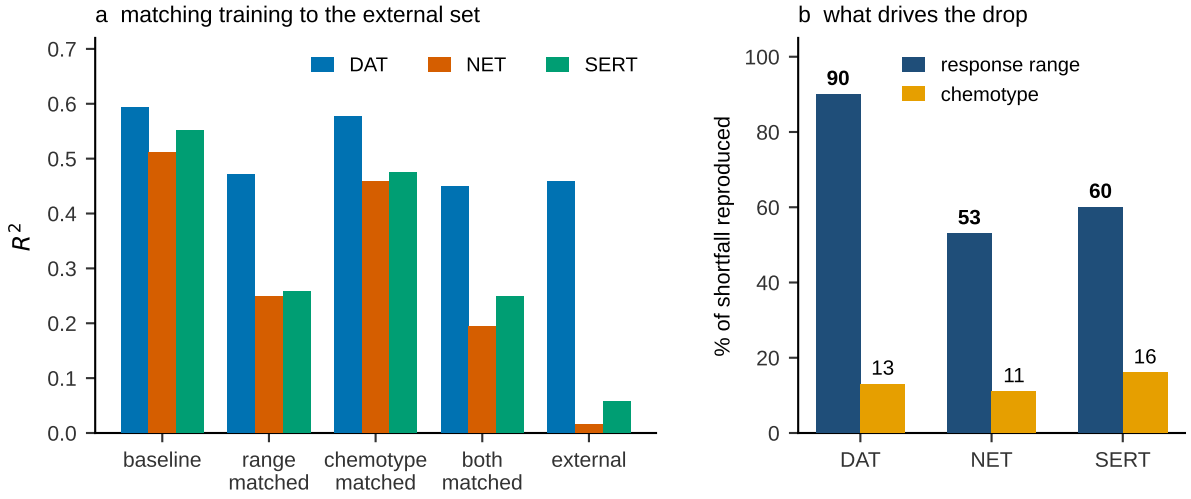


Figure 4: Decomposition of the external shortfall. (a) R^2 under five independent matched-training conditions; these are separate analyses, not a sequence. For DAT the both-matched internal estimate (0.450) and the external result (0.459) coincide. (b) Percentage of the shortfall reproduced by each matching strategy.

reproduces only 13%, 11% and 16%.

Applicability domain. Two complementary tests were applied. Mahalanobis distance — a multivariate measure accounting for covariance among descriptors, not a per-descriptor percentile check — placed the external compounds fully inside the training distribution (holdout median 2.61 versus training 2.73; 0% beyond the training 95th percentile of 4.81). By contrast, 40%, 33% and 64% of external compounds lay outside the training interquartile range at DAT, NET and SERT. Similarity to the training set is an established predictor of per-compound error. Sheridan tabulated it as a binned look-up populated by cross-validation at model-build time, and validated prospectively on five Merck assays and more than 6000 compounds measured *after* the models were built, finding the cross-validated bin error a semiquantitative predictor of the prospective error.Sheridan (2012) That study also reports, on its own data, that bins built on

Table 5: Error metrics across conditions. RMSE in log units.

Target	Baseline RMSE	Range-matched RMSE	External RMSE	Degradation
pDAT	0.732	0.668	1.174	+76%
pNET	0.807	0.680	0.848	+25%
pSERT	0.878	0.598	0.687	+15%

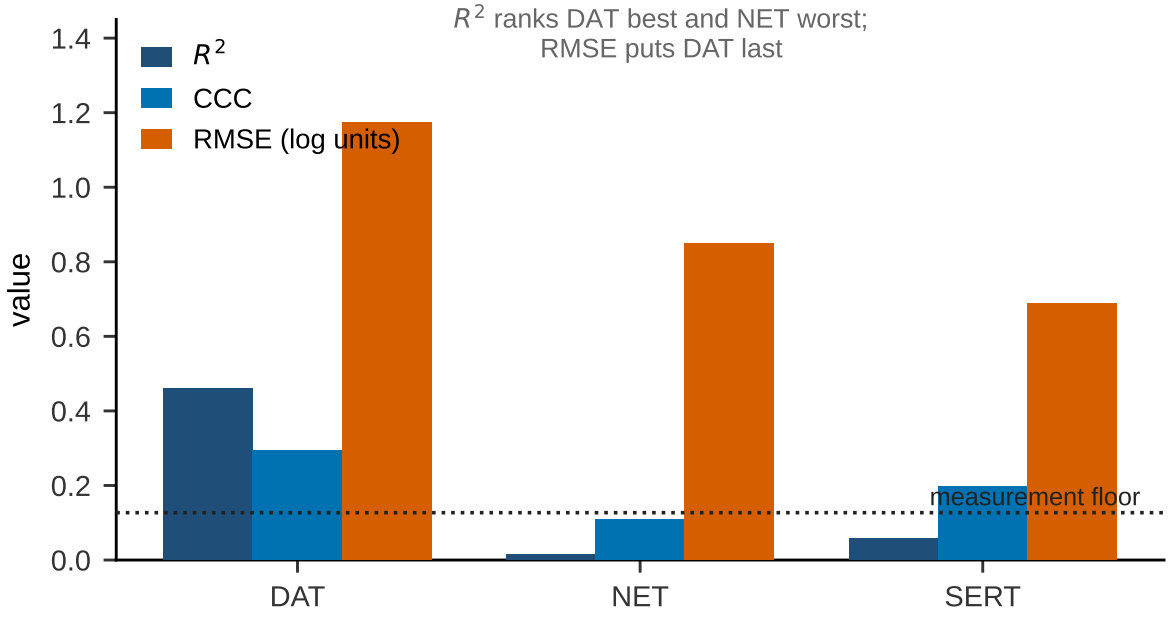


Figure 5: Metrics disagree on which target fared worst. R^2, concordance correlation coefficient and RMSE for the same external predictions. Dotted line is the aggregate measurement reproducibility floor (0.127 log units).

nearest-neighbour similarity alone cover a markedly smaller range of RMSE than bins built on prediction variance or on the predicted value — so similarity is the weakest of the three parameters it examined. The present result, that nearest-neighbour similarity contributes 2.2% uniquely once response-range position is in the model, is consistent with that ordering rather than at odds with it. Nearest-neighbour ECFP4 Tanimoto to training had a median of 0.505 with no compound above 0.85. The external compounds were therefore not extreme in the ten-descriptor Mahalanobis space and were not close fingerprint near-duplicates, although scaffold novelty was substantial and broader chemical novelty is excluded by neither test. Descriptor-space inclusion is not chemical familiarity.

On R^2, DAT appears best preserved and NET worst; on RMSE the order is close to reversed, DAT scoring worst and SERT best. That R^2 depends on the spread of the test response, because its reference is the training-set mean, was identified by Roy and co-workers, who proposed the r^2_m family as a remedy,Roy et al. (2009); Mitra et al. (2010) and has been restated with worked

examples by Alexander et al.Alexander et al. (2015) and by Martinez-Mayorga et al.,Martinez-Mayorga et al. (2024) the latter showing a case in which the model with the lowest absolute error also carries the lowest R^2 because its property range is narrower. Residual structure differs by target. DAT under-predicted throughout and worst at the potent end (observed minus predicted $+0.982$ overall, $+1.443$ in the more potent half); SERT over-predicted weak compounds (-0.737 in the weaker half). All three compressed toward the training mean: regressing prediction on observation gives slopes of 0.379, 0.076 and 0.157, against 1 for no compression (Figure 6).

The matching comparison cannot be run in the other direction. Table 4 matches the *training* set to the external set and asks how much of the shortfall that reproduces. The natural complement is to match the *test* set: draw internal test sets with the external set's size, response variance and nearest-neighbour similarity to training, and read off what the model would have scored on a test set of that shape. That comparison is the one most readers will want, and for most of these cells it does not exist.

For each external cell we asked where its response variance and its median nearest-neighbour Tanimoto fall within the distribution of values attainable by *any* random internal subset of the same size, over 20 000 draws (Table 6). For all nine cells at least one of the two lies outside that distribution, and in every cell it is the similarity: the external sets' median nearest-neighbour Tanimoto to training is 0.40–0.51, below the 1st percentile of what random internal subsets of the matching size attain (0.48–0.55), and in most cells below every one of the 20 000 draws. Response variance is unreachable in four of the nine cells in addition. Stalberga et al.'s SERT set has a response variance of 0.37 where fewer than 1% of internal subsets of 42 compounds fall below 1.07. No cell is matchable on both constraints, so the comparison is nowhere defined.

This is not a technicality about sampling. A rejection sampler with a widening tolerance — the obvious implementation — returns a band for all nine cells, because it widens until something is accepted, and every band it returns is matched to nothing. We ran four variants of this analysis during development and obtained three different headline counts across them (six, six, five and four of nine "inside the band") before establishing that no cell admits the comparison at all. The attainability check, not the band, is the result.

This failure also has an established name. Requiring that a matched comparator exist for every cell is the positivity, or common-support, condition of the matching and weighting literature,

and a sampler that returns an estimate where it is violated is the standard cautionary case there.Westreich and Cole (2010) We arrived at the check empirically and name it here so that the general theory, and its diagnostics, carry over: what we are reporting is that no internal subset lies in the support of the external sets' composition.

The one set that appeared matchable was the one that was contaminated. An earlier form of this analysis found three matchable cells, all three from the PDSP/BindingDB set, and the reason it did is itself a result. That set was then filtered for training overlap on the full InChIKey, which retains 30 of its 44 compounds. ECFP4 is computed without stereochemistry, so thirteen of those 30 differed from a training compound in stereochemistry alone and carried fingerprints identical to molecules the model had been fitted on: nine recorded without stereodescriptors where the training record specifies them, two resolved enantiomers of a racemate held flat in training, and two — the sertraline pair — resolved differently on both sides. Their nearest-neighbour similarity to training is 1.0 by construction, which lifted the set median to 0.63 and into the attainable range. The model's mean absolute error on those thirteen was 0.19–0.42 log units against 0.68–1.00 on the seventeen uncontaminated compounds. Filtering on the InChIKey skeleton block removes them, and the set becomes as unreachable as the other two. The attainability check selected, as the one external set comparable to internal data, the one external set that in part *was* internal data.

The two directions therefore answer different questions and give compatible answers. The training-side manipulation succeeds because the training set *contains* the external response region — one can restrict to it, and doing so reproduces 53–90% of the shortfall. The test-side manipulation fails because no random sample of the training set *resembles* the external sets — they are not a narrow slice of the training distribution but a deliberately selected region outside it, which is the quantitative form of the coverage result in § 3.11.

3.5 Assay modality (exploratory)

Kruskal–Wallis $p = 2.7 \times 10^{-11}$. Cross-assay dispersion is 2.6–3.4 times the replicate reproducibility floor at every target, making assay modality a larger error source than measurement error. Applying the measured NET offset as a constant correction did *not* improve external fit (RMSE 0.848 $\rightarrow$ 0.925). The dispersion of the offset (SD 0.658) is larger than its median

Table 6: Whether a composition-matched internal comparator exists. Percentiles are the position of the external value within the distribution attainable by random internal subsets of the same size (20 000 draws). "—" marks a value below the 1st percentile of that distribution, for which no matched comparator can be constructed. Every cell fails on similarity, and four also on variance; the sampler nonetheless returns a band for all nine. All three sets are filtered for training overlap on the InChIKey skeleton block; under the full-key filter used in an earlier draft the PDSP set retained 30 compounds, 13 of them fingerprint-identical to training compounds, and its similarity median rose to 0.63 (69th percentile).

Set	Target	n	Variance	Percentile	NN Tanimoto	Percentile
Stalberga 2026	DAT	42	0.77	—	0.51	—
	NET	42	0.60	—	0.51	—
	SERT	42	0.37	—	0.51	—
PDSP/BindingDB	DAT	17	1.08	30th	0.41	—
	NET	17	1.42	65th	0.41	—
	SERT	17	1.14	11th	0.41	—
Cui 2026	DAT	11	0.37	—	0.41	—
	NET	19	1.24	46th	0.40	—
	SERT	18	0.77	1st	0.40	—

Table 7: Binding-versus-functional offsets. Positive values indicate higher potency in the functional assay.

Target	n	Median offset	SD
DAT	112	−0.168	0.529
NET	97	**+0.440**	0.658
SERT	77	−0.083	0.497

(0.440), so the discrepancy is not a fixed additive shift but varies by compound and, plausibly, by assay conditions such as substrate concentration and incubation time, which differ between the contributing studies. A correction that transfers between datasets would need to be conditioned on those factors rather than applied as a constant.

3.6 The NET offset replicates and is not carried by one chemotype

Two questions follow from § 3.5. Does the offset survive a different retrieval of the source database, and is it a property of the transporter or of a single chemotype? Both were addressed with the independent re-retrieval described in § 2.11.

Replication. The NET offset reproduces at +0.540 (95% CI +0.390 to +0.670; $n = 81$; Wilcoxon $p = 1.2 \times 10^{-8}$), slightly larger than the frozen estimate of +0.440. DAT gives −0.280 ($[-0.380, -0.170]$, $n = 89$) and SERT −0.080 ($[-0.180, +0.000]$, $n = 71$); Kruskal–Wallis across the three targets $p = 2.7 \times 10^{-16}$. The NET-versus-other-targets contrast therefore holds across sixteen months of database revision and an independently written curation script.

Chemotype stratification. Every stratum of the re-retrieved NET set with $n \geq 5$ shows a

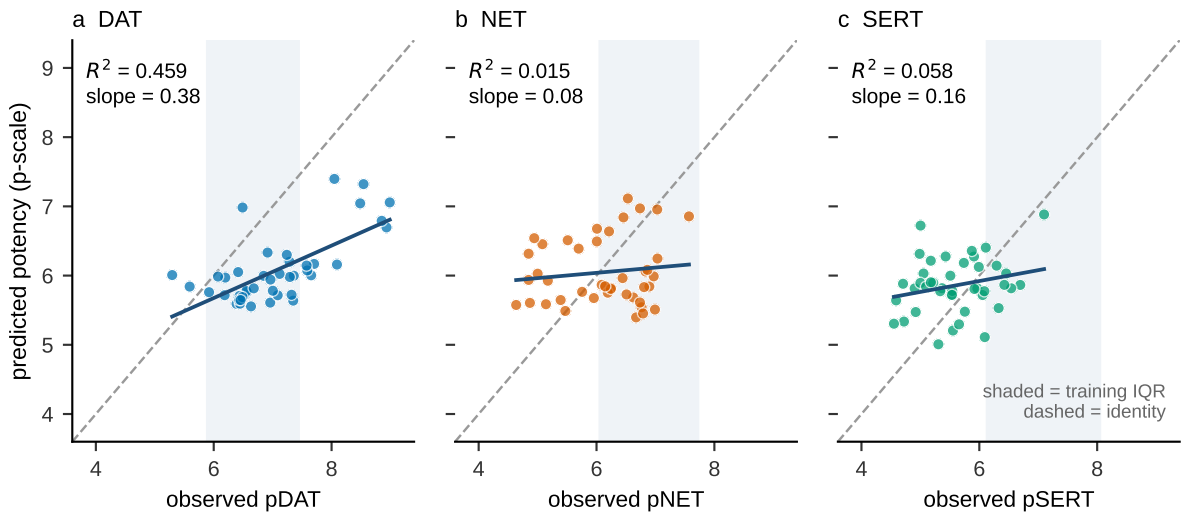


Figure 6: External predictions compress toward the training mean. Observed potency on the abscissa and predicted on the ordinate, so the solid line is the regression of prediction on observation and its slope is the annotated one; a slope of 1 would mean no compression. Dashed line is identity, and the line is drawn only across the observed range. Shaded band marks the training interquartile range.

positive median offset (Table 8, Figure 10): five of five Butina clusters at 0.6 Tanimoto, and four of four Bemis–Murcko scaffold strata, the latter spanning +0.31 to +1.07. Deleting the largest cluster outright — the 25 pyrrolidinophenones — leaves the pooled offset at +0.390; no leave-one-stratum-out deletion moves it by more than 0.15 or changes its sign. No single chemotype carries the effect. That is a statement about chemotypes as the manuscript's own strata define them; § 3.8 shows that the radioligand strata, which are close to disjoint in chemistry, do not behave alike, and that the chemotype strata used here do not settle the question.

But the magnitude is chemotype-dependent. The offset differs significantly across strata (Kruskal–Wallis, $p < 0.001$), spanning roughly threefold from +0.31 to +0.99, and is largest in the pyrrolidinophenone (α-pyrrolidinovalerophenone-type) cathinones — the chemotype of greatest interest for novel psychoactive substance profiling. The pooled figure is thus an average over a heterogeneous set. This is consistent with, and supplies a mechanism for, the failure of the constant correction reported in § 3.5: a transferable correction would have to be chemotype-resolved, and the per-stratum samples available here ($n = 5$ to $n = 25$) are too small to estimate one.

One discrepancy between retrievals. In the frozen analysis the DAT and SERT offsets were not distinguishable from one another ($p = 0.39$). In the re-retrieval they are (-0.280 versus -0.080; Mann–Whitney $p = 0.005$), driven by a more negative DAT estimate on a

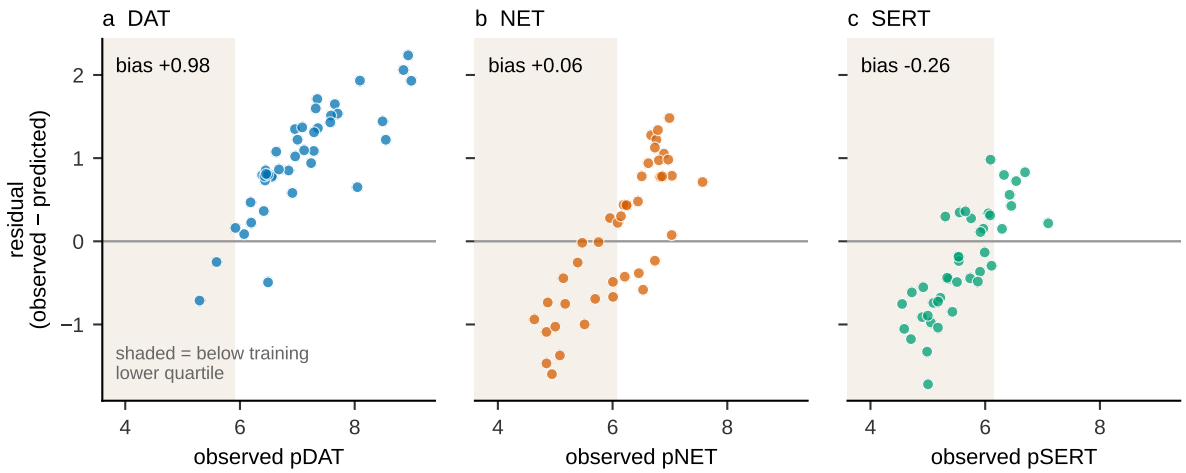


Figure 7: Residual structure differs by target. Shaded region marks values below the training lower quartile. DAT under-predicts its most potent compounds; SERT over-predicts weak ones.

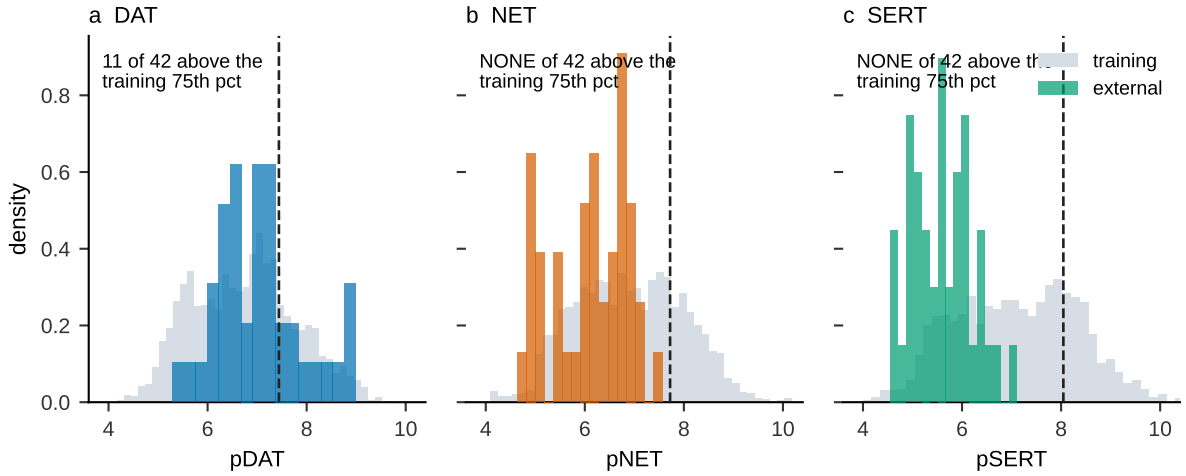


Figure 8: The advance-specified external set contains no potent NET or SERT ligands. Grey is the uncensored training distribution and the dashed line its 75th percentile; the coloured histogram is the 42 external compounds. Eleven clear the line at DAT and none at NET or SERT. The two sets assembled afterwards do reach above it (Figure 13).

different compound set. We therefore claim that the offset is specific to NET among the three targets — a contrast robust to retrieval and curation — and *not* that DAT and SERT behave identically. The caption of Figure 9 should be read in that narrower sense.

3.7 The offset survives within single papers and is not the releaser effect

It is not a between-laboratory artefact. Of the 81 NET paired compounds, 75 have both their binding K_i and their functional IC_{50} reported in the *same* paper. Restricted to those within-document pairs the offset is $+0.550$ (95% CI $[+0.400, +0.690]$; Wilcoxon $p = 1.0 \times 10^{-9}$), indistinguishable from the $+0.540$ obtained by pooling across documents. Five documents

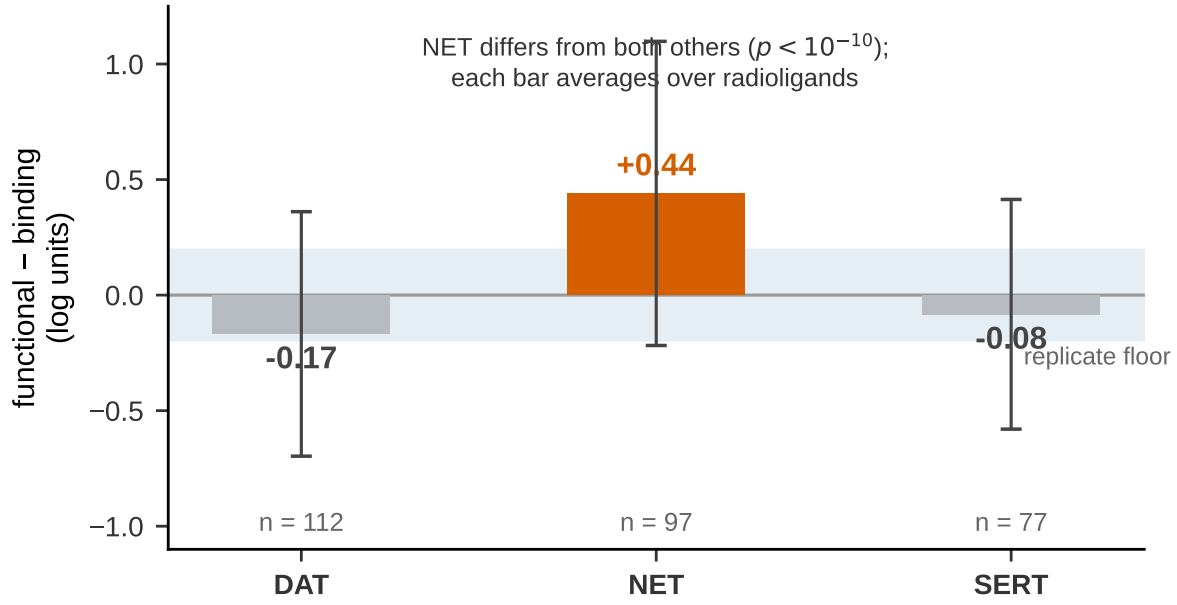


Figure 9: The binding-versus-functional offset is positive at NET and not at DAT or SERT, on the frozen set. Median difference between functional and binding potency for compounds measured both ways, with SD; shaded band is the replicate reproducibility floor. The robust contrast is NET against the other two. DAT and SERT are not distinguishable from one another here ($p = 0.39$) but are on the independent re-retrieval ($p = 0.005$), so no claim of DAT–SERT equality should be read from this figure. Each bar also averages over radioligands, which Table 10 separates.

contribute, from five different laboratories and spanning 1999–2007 (Figure 11): Choi et al. 1999 Choi et al. (1999) ($n = 10$, +0.15), Houlihan et al. 2002 Houlihan et al. (2002) ($n = 13$, +1.12), Axford et al. 2003 Axford et al. (2003) ($n = 8$, +0.47), Meltzer et al. 2006 Meltzer et al. (2006) ($n = 29$, +0.99) and Gong et al. 2007 Gong et al. (2007) ($n = 17$, +0.31). Every one is positive. Dropping the largest contributor leaves a median of +0.390 across the remaining four documents and 48 compounds. Because the two values within a paper are produced by one group on one compound batch, this comparison removes between-laboratory calibration differences entirely,

Table 8: Chemotype stratification of the NET binding-versus-functional offset on the independent re-retrieval ($n = 81$). Strata are Butina clusters of ECFP4 fingerprints at 0.6 Tanimoto. Stratum labels are descriptive readings of the Bemis–Murcko scaffold of each cluster centroid; the exact scaffold SMILES are in the deposited repository. Confidence intervals are percentile bootstrap (4000 resamples). Every stratum is positive.

Chemotype stratum	n	Median offset	95% CI
Pyrrolidinophenones (α-PVP-type)	25	+0.99	[+0.69, +1.10]
Fused imidazoline tricyclics	10	+0.86	[+0.38, +1.38]
Naphthyl norbornanes	5	+0.64	[+0.39, +0.78]
Benzhydryl ether amines (GBR-type)	5	+0.40	[−0.15, +0.45]
3-Aryltropanes	17	+0.31	[−0.17, +0.39]
All compounds	81	**+0.54**	[+0.39, +0.67]

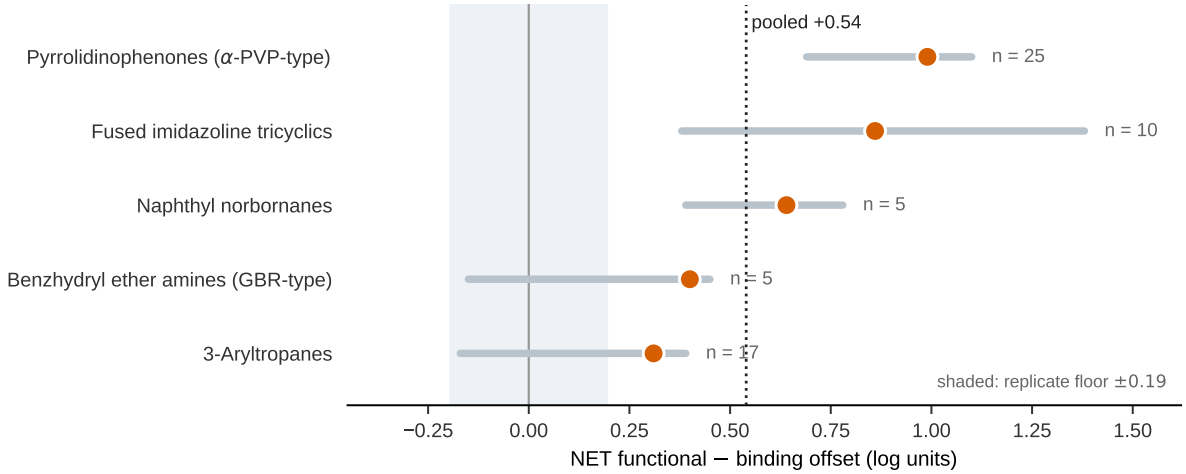


Figure 10: The NET binding-versus-functional offset within each chemotype, on the independent re-retrieval. Every stratum is positive; the magnitude varies roughly threefold. Dotted line marks the pooled estimate. Bars are 95% percentile bootstrap intervals.

and the offset is unchanged.

The same test at DAT and SERT separates NET from the other two. Repeating the analysis on the DAT and SERT paired sets gives -0.280 on 86 within-document pairs (95% CI $[-0.380, -0.155]$) and -0.090 on 66 (95% CI $[-0.220, -0.010]$), against $+0.550$ at NET; Kruskal–Wallis across the three $H = 77.6$, $p = 1.4 \times 10^{-17}$. Adding the Eshleman blockers described below as a sixth source — merging the two compounds common to both, cocaine and mazindol, so that no compound is counted twice — gives pooled estimates of -0.180 at DAT ($n = 100$, 95% CI $[-0.300, -0.070]$), $+\mathbf{0.600}$ at NET ($n = 89$, $[+0.470, +0.748]$, Wilcoxon $p = 2.9 \times 10^{-12}$) and -0.048 at SERT ($n = 80$, $[-0.130, +0.110]$, $p = 0.38$); Kruskal–Wallis $H = 91.5$, $p = 1.3 \times 10^{-20}$. Only NET is distinguishable from zero in the positive direction, and SERT is not distinguishable from zero at all.

The per-source pattern is the more informative one. At NET all six sources agree in sign; at DAT and SERT they do not (DAT -0.51 to $+0.28$, SERT -0.34 to $+0.19$), so what is a consistent directional offset at NET is scatter about zero at the other two transporters (Figure 11). This is the same NET-specific contrast reported in § 3.6 and Table 7, now established without any between-study comparison.

Robustness of the NET result. Four distribution-free checks, each aimed at a different objection. (i) The direction is near-unanimous at the compound level rather than carried by a

tail: 64 of 75 within-document NET pairs are positive (85.3%, exact 95% CI [75.3, 92.4], sign test $p = 3.1 \times 10^{-10}$), against 24 of 86 at DAT and 24 of 66 at SERT. (ii) A sign-flip permutation test — the exact null for a paired design, in which the two modality labels are exchangeable within a compound — gives an observed median of +0.550 against a null range of $[-0.450, +0.400]$ over 20 000 relabellings ($p < 5 \times 10^{-5}$); the corresponding SERT value falls inside its null ($p = 0.21$). (iii) Taking the *source* rather than the compound as the unit of analysis, 6 of 6 sources are positive at NET (sign test $p = 0.031$, the minimum attainable at $n = 6$) against 2 of 6 at DAT and 2 of 5 at SERT. (iv) The separation between targets is large by distribution-free effect size: Cliff's $\delta = +0.729$ for NET against DAT and +0.665 against SERT. Leave-one-document-out leaves +0.390 at worst, removing Meltzer et al.'s 29 compounds.

It is not the releaser-versus-blocker distinction. All 81 paired compounds classify as uptake blockers. None carries a release or efflux assay record in ChEMBL (0 of 785 retrieved DAT/NET/SERT records), and none matches the substrate-type releaser pharmacophore: the set is dominated by α-pyrrolidinophenones, mazindol-type isoindoles, 3β-aryltropanes and benzhydryl ethers, with a median molecular weight of 303 Da and a single compound below 230 Da. Amphetamine-type releasers are absent by construction, because paired binding-and-functional measurements at NET come almost entirely from medicinal-chemistry uptake-inhibitor series. Two internal controls confirm the scale: cocaine gives +0.78 and mazindol +0.50 here, against the 7.0- and 5.0-fold NET ratios (+0.85 and +0.70 log units) reported for the same two compounds by Eshleman et al.Eshleman et al. (2017) The offset is therefore a property of blockers, measured within blockers, and the releaser mechanism cannot account for it.

Half of the paired offset is the value type, not the assay format. The +0.550 above compares binding K_i with functional IC_{50}, and in ChEMBL those two attributes travel together: binding records are predominantly K_i and functional records predominantly IC_{50}. The comparison therefore moves assay format and value type at once, and Cheng–Prusoff alone predicts a difference between K_i and IC_{50}.Kalliokoski et al. (2013) With modality labelled from assay text rather than from `standard_type` (§ 2.2) the off-diagonal cells are populated — 3018 binding IC_{50} and 1336 functional K_i records — and the two can be estimated in one model.

Fitted with compound-within-document fixed effects and document-clustered standard errors across 1169 informative records from 49 documents and 512 compounds, the assay-format com-

ponent at NET is +**0.261** (95% CI [+0.06, +0.46], $p = 0.010$) and the K_i-versus-IC_{50} component is indistinguishable from zero (-0.047, $p = 0.52$). Both components are null at DAT (+0.003, $p = 0.98$; value type +0.107, $p = 0.088$) and SERT (+0.036, $p = 0.84$; value type +0.184, $p = 0.20$), so the NET specificity reported above survives the decomposition while its magnitude halves (Table 9).

The two estimates are not in conflict; the larger one is the sum of both components. Applied to the five paired documents alone, the same model gives +0.578 for format when the value-type term is omitted — recovering the +0.550 reported above — and +0.289 when it is included, which is within confidence of the corpus-wide +0.261. The paired estimate is the clean within-series demonstration; the fixed-effects estimate is the one to generalise from, and it rests on roughly eight times as many documents.

The format effect survives among blockers alone, at corpus scale. The structural classification of § 2.3 assigns 42 of 5796 compounds as substrate-type and 5754 as blockers. Restricting the model to blockers leaves the NET format effect at +0.238 (95% CI [+0.04, +0.43], $p = 0.018$; 1095 records, 49 documents, 487 compounds), and substrate-type compounds are 1.9% of the NET informative records, so they cannot be carrying it. The substrate stratum itself shows a larger offset (+1.28), directionally consistent with the $K_m \neq K_D$ account Schömig et al. (1988) and with the Eshleman releaser values above, but it rests on three documents and no interval is quoted for it.

Robustness: leave-one-document-out leaves the corpus-wide NET estimate between +0.227 and +0.331 and significant in 272 of 272 fits; moving the `conflict` class into either modality leaves it at +0.261, since only 40 of those 810 records are NET and none falls in an informative group; document fixed effects in place of compound-within-document effects give +0.253 ($p = 0.032$).

The cluster count. Forty-nine clusters is few enough that CR0 cluster-robust variance over-rejects, so the interval above is not the one to quote. A wild cluster bootstrap-t with Rademacher weights and 9999 replications, clustered on document and imposing the null, widens the NET format effect to [+0.024, +0.486] with $p = 0.028$ against the CR0 [+0.062, +0.460] and $p = 0.010$. The correction matters and the effect survives it. The value-type component is null either way ([−0.17, +0.07], $p = 0.53$). Intervals elsewhere in this paper are CR0 unless stated, including in Table 10, where quoting ten bootstrap intervals would obscure the comparison between strata;

Table 9: Assay format and value type, estimated jointly. Compound-within-document fixed effects, standard errors clustered on document. Positive values mean the first-named condition reads more potent. The format effect is NET-specific; the value-type effect is not, and at NET it is absent.

Target	Records	Documents	Compounds	Functional − binding	pK_i − pIC_{50}
DAT	932	49	388	$+0.003$ $[-0.16, +0.17]$	$+0.107$ $[-0.02, +0.23]$
NET	1169	49	512	$+\mathbf{0.261}$ $[+0.06, +0.46]$	-0.047 $[-0.19, +0.09]$
SERT	939	47	380	$+0.036$ $[-0.32, +0.39]$	$+0.184$ $[-0.10, +0.46]$
NET, uptake blockers only				$+0.238$ $[+0.04, +0.43]$	

the bootstrap interval is given in the text for each estimate the argument rests on, and is always the wider of the two.

3.8 The offset is a property of the assay cell, not of the assay format

Everything above treats the binding side of the comparison as one thing. It is not. The assay descriptions name the radioligand (§ 2.4), and splitting on it changes the reading of the result.

At NET the effect is confined to one radioligand. Refitting the model of Table 9 inside each radioligand stratum — still compound-within-document effects, still document-clustered — gives a format effect of $+\mathbf{0.507}$ (CR0 95% CI $[+0.34, +0.68]$, $p = 3 \times 10^{-9}$; wild cluster bootstrap-t $[+0.20, +0.82]$, $p = 0.005$, and the bootstrap is the interval to read, on 17 clusters) in the 17 documents using [^{125}I]RTI-55, and $+\mathbf{0.038}$ ($[-0.18, +0.26]$, $p = 0.73$) in the 21 using [^{3}H]nisoxetine (Table 10). The pooled $+0.261$ is an average of a large effect and a null; the nisoxetine interval is centred near zero and only barely reaches the pooled estimate.

The within-document medians agree, which matters because that contrast removes between-laboratory calibration by construction: $+0.530$ over 175 pairs in 12 RTI-55 documents against -0.115 over 146 pairs in 10 nisoxetine documents (Mann–Whitney $p = 5.5 \times 10^{-16}$). Eleven of the twelve RTI-55 documents are positive; the ten nisoxetine documents are six negative and four positive, centred near zero. Four of the five papers behind the $+0.550$ paired estimate of § 3.7 use [^{125}I]RTI-55.

It is not simply that RTI-55 reads low. Within the RTI-55 stratum NET remains the outlier, at $+0.507$ against $+0.183$ at DAT and -0.031 at SERT. And within the target-specific ligands the spread is wider still: -0.308 for [^{3}H]WIN 35,428 at DAT and -1.044 for [^{3}H]paroxetine

Table 10: The binding-versus-functional offset by binding radioligand. Compound-within-document fixed effects, document-clustered standard errors, as in Table 9; a document is assigned to the radioligand it uses for the most binding records. The *all* rows are Table 9 repeated for comparison. The format effect is not a property of the format: it spans 1.5 log units across binding assays.

Target	Binding radioligand	Documents	Records	Functional − binding
DAT	all	49	932	+0.003 [−0.16, +0.17]
	[^{125}I]RTI-55	20	634	+0.183 [−0.01, +0.38]
	[^{3}H]WIN 35,428	12	148	−0.308 [−0.49, −0.13]
NET	all	49	1169	**+0.261** [+0.06, +0.46]
	[^{125}I]RTI-55	17	654	**+0.507** [+0.34, +0.68]
	[^{3}H]nisoxetine	21	411	**+0.038** [−0.18, +0.26]
SERT	all	47	939	+0.036 [−0.32, +0.39]
	[^{125}I]RTI-55	18	417	−0.031 [−0.22, +0.16]
	[^{3}H]citalopram	9	80	−0.055 [−0.21, +0.10]
	[^{3}H]paroxetine	6	274	−1.044 [−1.39, −0.70]

at SERT. The offset is therefore a property of the assay cell — here indexed by target and radioligand, which is the stratification the data support and not a claim about which of the two is operative — ranging over 1.5 log units across the binding assays in this corpus, and each per-target estimate is a composition-weighted average over whichever radioligands happen to dominate that target's literature. NET looks special in pooled data because NET is where [^{125}I]RTI-55 — the cell with the largest positive offset — is most used.

What this does and does not establish. Radioligand is confounded with document and nearly with chemotype, and the data cannot separate them. Only four NET documents use more than one binding radioligand and none has enough two-ligand overlap to give a within-document radioligand contrast. The two NET strata share 5 compounds out of 174 and 163, their median molecular weights are 285 and 365, and their median cross-stratum nearest-neighbour Tanimoto is 0.28 — the RTI-55 stratum is stimulant-like and the nisoxetine stratum is medicinal-chemistry NET series. What the stratification establishes is that the pooled estimate is an average over heterogeneous assay cells, not that the radioligand rather than the chemistry is the operative variable. A mis-specified K_D in the Cheng–Prusoff conversion for RTI-55 at NET would produce exactly a constant shift in every K_i from that assay at one target, which fits the observed pattern, but this corpus cannot confirm it. Resolving it needs an experiment: the same compounds at NET by both radioligands in one laboratory.

The offset is real and not exploitable. Separating the two modalities does not improve prediction. Trained and cross-validated on modality-specific labels, models predicting modality-specific values gain $+0.002$ r^2 on average over models trained on pooled labels (range -0.010 to $+0.016$ across six target $\times$ modality cells). Externally, where each test set was measured in one modality — Stalberga et al. and Cui et al. by functional uptake, the PDSP/BindingDB panel by radioligand binding — scoring with the matching-modality model is *worse* than scoring with the pooled model by -0.067 r^2 (better in 2 of 9 cells, Wilcoxon $p = 0.25$). A size-matched control attributes most of that to data quantity rather than to modality: refitting the pooled model on a random subset the size of the modality model costs -0.050 on its own, leaving -0.017 attributable to the separation itself.

This closes the loop on the constant correction reported in § 3.5, which also worsened external fit. The offset at NET is measurable, replicable, specific and mechanistically interpretable, and none of the three obvious ways of acting on it — correcting by it, training separately on it, or selecting the matching modality at test time — recovers any accuracy. A difference that is detectable in aggregate need not be large enough, relative to the spread around it (SD 0.658 against a median of 0.440), to carry information about any individual compound.

An independent within-document replication, with the classes separated. Eshleman et al.Eshleman et al. (2017) is itself a single-laboratory source reporting both modalities on the same compounds, and it classifies each compound as blocker or substrate by a [^{3}H]neurotransmitter *release* assay — a criterion that does not use the ratio under test. Their Table 2 therefore permits the cleanest available version of this comparison. Across their 16 uptake blockers the offset is $+0.752$ at NET (95% CI $[+0.699, +0.914]$) against $+0.279$ at DAT and $+0.176$ at SERT (Kruskal–Wallis $H = 26.0$, $p = 2.2 \times 10^{-6}$) — the same NET-specific excess, in an independent dataset, among compounds independently confirmed not to be substrates. Their 15 substrate-type releasers sit one to two log units higher at every transporter ($+1.36$, $+2.14$, $+1.54$ at DAT, NET, SERT), which is why the class has to be excluded before the modality effect can be read, and why a ratio-based classification of the present paired set would have been circular. The elevation is expected and is not a competing measurement of the same quantity: substrate affinity is low by design, because a transported compound must release on the far side of the membrane, while a substrate suppresses *net* uptake not only by competing for the site but by consuming transport cycles, eroding the sodium gradient and driving reverse transport

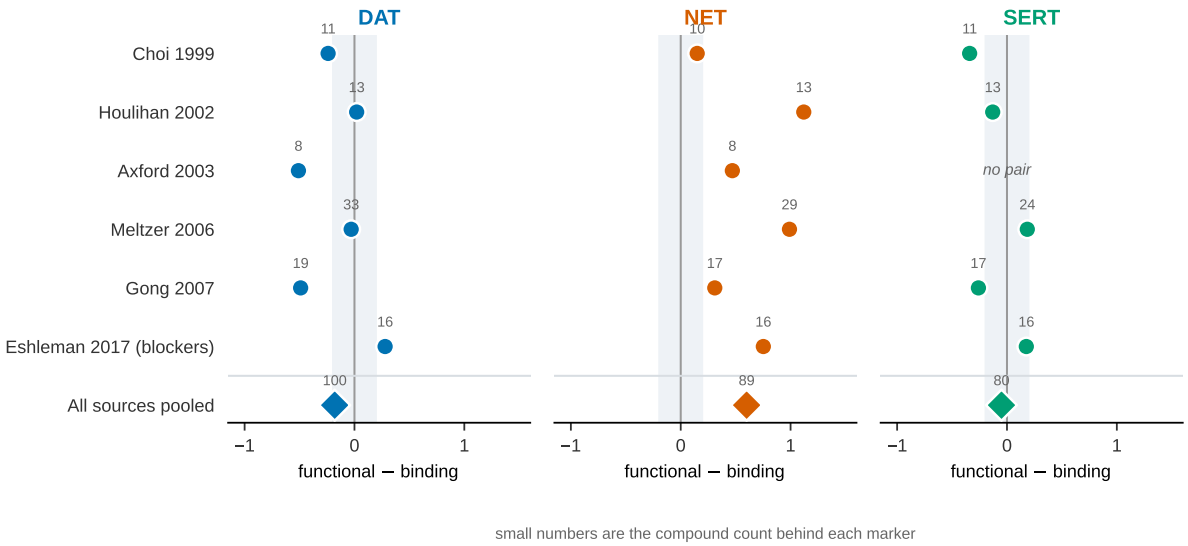


Figure 11: The binding-versus-functional offset measured within single papers, where both modalities come from one laboratory on one compound series, so no between-study calibration difference can contribute. Each of the first six rows is one source; the final row pools them, taking one value per compound so that cocaine and mazindol, which appear in both the Eshleman panel and the ChEMBL series, are counted once. Markers are medians and bars 95% percentile bootstrap intervals. At NET every source agrees in sign; at DAT and SERT they do not. Small numbers are the compound count behind each marker. Axford et al. contributed no SERT pair. The Eshleman row is their uptake blockers only, classified by their release assay; their substrate-type releasers are excluded and are reported in the text.

— a large K_i and a small IC_{50} from the same molecule. Notably the shift is close to uniform across the three transporters (+1.08, +1.39, +1.36 relative to the same paper's blockers), so the NET-specific excess is approximately additive on top of it (+0.53 within blockers, +0.69 within releasers) rather than being produced by it.

3.9 Extended external evaluation (exploratory)

The extended set comprises 114 compounds across 31 scaffolds from four laboratories, with response range restored to near-training levels (log-ratio SD 1.39 versus 1.41). Document-level leakage probes returned zero DAT/NET/SERT activities from any contributing source. The log-ratio is modelled *directly*, as one regressor trained on the log-ratio, rather than differenced from the two per-target models; the differenced route is less conservative ($R^2 = 0.352$ on the same compounds). The pooled statistic is computed on the 107 compounds from the three Basel and Linköping sources, the seven Eshleman compounds being reported separately by source in Table 11 rather than pooled. DAT/SERT log-ratio prediction: $R^2 = 0.183$, 95% CI $[0.058, 0.336]$, $r = +0.427$, $p = 4.4 \times 10^{-6}$, RMSE 1.335, CCC 0.304.

Two features of Table 11 caution against reading it as a laboratory effect. The sources differ

Table 11: Extended external performance by source. Scaf. = distinct Murcko scaffolds; NN = nearest-neighbour Tanimoto to training. Confidence intervals are percentile bootstrap (2000 resamples) and overlap substantially across sources. The OHSU row ($n = 7$, 3 scaffolds) is reported for completeness only: its R^2 is not interpretable at that sample size, as its simultaneously highest R^2 and worst RMSE illustrate.

Source	n	Scaf.	SD(y)	R^2 (95% CI)	RMSE	MAE	Bias	Med NN
Luethi & Liechti 2020 (Basel)	39	12	1.33	0.412 [0.143, 0.649]	1.065	0.842	−0.309	0.500
Basel primary studies	26	11	1.15	0.357 [0.084, 0.655]	1.068	0.815	+0.565	0.444
Stalberga et al. 2026 (Linköping)	42	14	0.97	0.114 [0.001, 0.420]	1.668	1.463	+1.383	0.505
Eshleman et al. 2017 (OHSU)[a]	7	3	0.78	0.773 [0.446, 0.991]	1.687	1.651	+1.651	0.667

Table 12: Variance decomposition of external error. Unique contribution is the semi-partial R^2 of an ordinary least-squares model of absolute prediction error: the drop in model R^2 when that block alone is removed. Confidence intervals are percentile bootstrap over external compounds (2000 resamples).

Predictor block	R^2 alone	**Unique**	95% CI
Response-range position	0.444	**0.308**	[0.187, 0.416]
Chemical similarity	0.024	0.022	[0.000, 0.071]
Source identity	0.178	0.021	[0.005, 0.077]
All three together	**0.495**	—	[0.374, 0.639]

significantly in chemical novelty (Kruskal–Wallis, $H = 13.95$, $p = 0.003$), so they cannot be treated as comparably novel. And the OHSU set has the highest R^2 (0.773) together with the worst RMSE (1.687) and largest bias (+1.651) — at $n = 7$, R^2 is not interpretable.

This diagnostic is largely circular. The failure has a name outside cheminformatics: it is endogenous selection bias from conditioning on a quantity that is itself a function of the outcome, the general statement of which is the causal-inference literature on collider and post-outcome conditioning.Elwert and Winship (2014) Response-range position is computed from the observed external value, and absolute error is computed from the same value. A regularised model shrinks predictions toward the training mean, so error is mechanically larger for compounds far from that mean. Here predictions had SD 0.717 against observations at 1.382, and an observation's distance beyond the training interquartile range correlated with its distance from the training mean at $r = 0.983$.

Three comparisons establish that the apparent effect is mechanical (Table 13). Regressing absolute error on range position gives $R^2 = 0.444$. Substituting the naive shrinkage predictor — distance from the training mean, which involves no notion of range — reproduces it at $R^2 = 0.448$. The only form available before the external outcome is known, using the distance of the *prediction* from the training mean, collapses to $R^2 = 0.001$. A model predicting the training mean for every compound would give $R^2 = 0.966$ by construction.

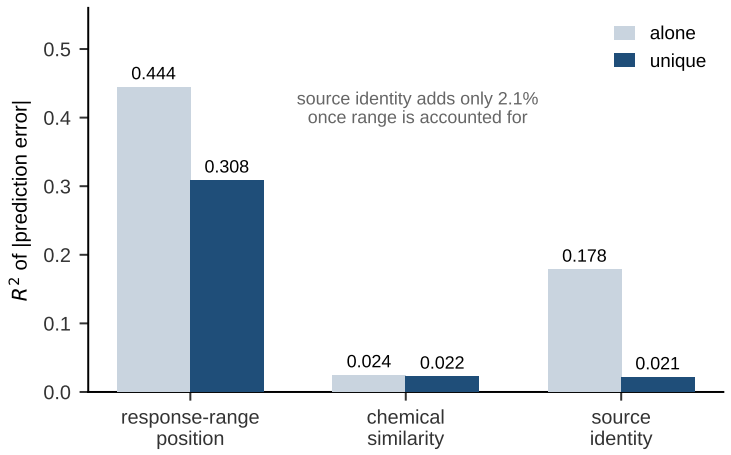


Figure 12: An outcome-conditioned error diagnostic on the extended external set ($n = 114$). Unique contributions from a regression of absolute prediction error on response-range position, chemical similarity and source identity; "alone" is each block's own R^2 and "unique" the drop in model R^2 when that block is removed. The figure is reported as a caution rather than as a result: the analysis is outcome-conditioned and largely reproduces prediction shrinkage (Table 13).

Table 13: An outcome-conditioned error diagnostic is largely circular. Each row regresses absolute external error on a single predictor. The quantity available before the external outcome is known (third row) carries essentially no information.

Predictor of |external error|	R^2
Response-range position (outcome-conditioned)	0.444
Distance from training mean (shrinkage null)	0.448
Distance of *prediction* from training mean	**0.001**
Mechanical ceiling (model predicts training mean)	0.966

We therefore make no causal claim about the determinants of external error. What the analysis establishes is narrower and still useful: external error here is dominated by prediction shrinkage outside the training response region, a property of regularised regression rather than a discovery about chemical generalisation. Investigators reporting similar decompositions should check them against a shrinkage null before interpretation. Source identity and chemical similarity contribute little unique variance conditional on the response-range measure included here; because the two predictors are strongly confounded by construction, that is a statement about this regression and not evidence that laboratory effects are unimportant in general.

What to use instead. Ending on a caution leaves the practical question open, and it has an answer this paper did not supply. The quantity the failed diagnostic was reaching for — a per-compound estimate of how far a prediction can be trusted, available *before* the outcome is known — is what conformal prediction provides, with a validity guarantee under exchangeability and without any conditioning on the observed value.Norinder et al. (2014) A Mondrian variant

stratified on predicted response region would address precisely the heterogeneity this section identifies, since the guarantee then holds within stratum rather than only on average. We have not fitted one here, and the benchmark's exchangeability assumption would itself need checking against the coverage results of § 3.11 — an external set drawn from a different response region is not exchangeable with the training set, which is the same objection in a different vocabulary. That is the natural next step on this benchmark and we flag it as the recommendation the negative result points to.

3.10 Familiar chemistry, independent measurement (pilot)

Table 14: Familiar chemistry, independent measurement ($n = 4$).

Comparison	RMSE (log units)
Model prediction vs independent measurement	0.878
Model prediction vs its own training value	0.277
Training value vs independent measurement	**0.683**

The third row involves no modelling: it is disagreement between two laboratories measuring the same molecules. At 0.683 log units it approaches the model's own external error. At $n = 4$ this is not a result, but it is a design worth executing at scale.

3.11 Why the advance-specified test could not evaluate NET or SERT

None of the 42 compounds in the advance-specified external set exceeded the training 75th percentile at either NET or SERT, against eleven that did at DAT; external SERT values had a median of 5.54 against an uncensored training median of 7.13. The 400 most SERT-potent training compounds are dominated by tropanes, diaryl sulfides, aryloxy-pyrrolidines and tetralins — the SSRI and tricyclic chemotypes — and none of these scaffolds appears in any published external profiling set we could identify. External characterisation of these transporters overwhelmingly concerns DAT-directed stimulants, which are by design weak at NET and SERT (Figure 8).

The consequence for R^2. This is not only a question of statistical power. R^2 is normalised by the variance of the set on which it is computed, so a test set that occupies a narrow slice of the response range carries a lower achievable R^2 for any model whatever. On the advance-specified external compounds the variance is 0.70 of the uncensored training variance at DAT

but 0.51 at NET and 0.25 at SERT, and the median external compound sits at the 57th, 30th and 12th training percentile respectively. Taking the cross-assay dispersion measured on the independent re-retrieval of § 2.11 as the applicable noise term (SD 0.445, 0.655 and 0.492 over 89, 81 and 71 compounds measured both ways; the external values are uptake IC_{50} against a binding-dominated training median, so the modality offset applies in full), a model predicting the true pooled value perfectly would score $R^2 = 1 - \sigma_e^2/\sigma_y^2$ of 0.741 at DAT, 0.279 at NET and 0.354 at SERT on this set — against 0.820, 0.633 and 0.838 on the training distribution. The achieved values are thus 62%, 5% and 16% of what was reachable (Table 15).

Two external sets assembled after the advance-specified analysis, described in § 3.12, reach the part of the response range the stimulant panel cannot, and the ceiling recovers with the spread: at NET it rises from 0.279 to 0.853 and 0.655. For PDSP the matched noise term of Table 15 contributes to that as well as the spread does, since that set is scored in its own modality. The achieved fraction rises with it, from 5% to 21% and 52%, though only the second of those is distinguishable from zero. The advance-specified test's shortfall at NET and SERT is therefore at least partly a property of the only test set then available; how much of it is, rests on a single 20-compound series.

3.12 Two external sets assembled after the advance-specified analysis (exploratory)

The preceding section describes the external data available when the protocol was fixed. Two further sources were assembled afterwards and are reported here as an exploratory extension; neither was pre-specified.

PDSP/BindingDB. The NIMH Psychoactive Drug Screening Program's K_i database, as redistributed by BindingDB, yields 44 compounds with a value at all three transporters. BindingDB ingests only rows mappable to a UniProt accession, and Q01959, P23975 and P31645 are human-specific, so species is guaranteed structurally rather than parsed from free text; all retrieved rows are K_i, matching the binding-dominated training modality. Twenty-seven compounds were removed for training overlap on the InChIKey skeleton block, leaving 17, drawn from five source documents.

This set is the one place where the choice of overlap filter changes a reported result, and it

is worth being explicit about why. Removing overlaps on the *full* InChIKey — the advance-specified rule, and the rule applied to the advance-specified panel, where it makes no difference — leaves 30 compounds. Thirteen of those 30 differ from a training compound in stereochemistry alone: nine are recorded without stereodescriptors where the training record specifies them, two are resolved enantiomers of a racemate held flat in training, and two are the sertraline pair. Because the model's fingerprint is computed without stereochemistry, all thirteen are numerically indistinguishable from compounds it was fitted on, and the model reproduces them to a mean absolute error of 0.19–0.42 log units against 0.68–1.00 on the other seventeen. Retaining them raises this set's apparent SERT R^2 from 0.017 to 0.352 and its median nearest-neighbour similarity from 0.41 to 0.63, the latter being what made it the only external set with a computable matched comparator (Table 6). All figures reported here use the skeleton block. Two structures in the source file also carry a CXSMILES annotation block; stripping it before parsing is necessary for one of them, maprotiline, whose annotation the default parser rejects outright, so earlier builds dropped it without a warning; the other row parses either way. Readers comparing against the preprint should note that its PDSP figures were computed under the looser rule, and on 29 rather than 30 compounds for that reason.

Most of the literature behind this slice is absent from ChEMBL_37, because the measurements appear in *J Pharmacol Exp Ther* and *Eur J Pharmacol*, which ChEMBL curates only sparsely.

Cui et al. 2026. A 20-compound *N*-aryl-*N*-acyl diamine series developed as selective SERT and NET inhibitors.Cui et al. (2026) Potencies are uptake IC_{50} values at human SERT, NET and DAT stably expressed in HEK293 cells. The paper postdates the ChEMBL_37 literature freeze and is absent from that release (`document.json?doi` returns zero), so it is external by construction. It is also the most structurally independent test applied here: none of its ten Murcko scaffolds occurs in the training set, no compound matches a training InChIKey, and the median nearest-neighbour Tanimoto to training is 0.40.

Result. Both sets span response ranges the stimulant panel does not (Figure 13), and on one of them the NET and SERT correlations are significant where the advance-specified test could not distinguish them from zero (Table 15). On the Cui series the model, which has seen none of its ten Murcko scaffolds, reaches $R^2 = 0.342$ at NET ($p = 0.0085$) and 0.468 at SERT ($p = 0.0018$), against 0.015 ($p = 0.44$) and 0.058 ($p = 0.12$) on the advance-specified panel, with

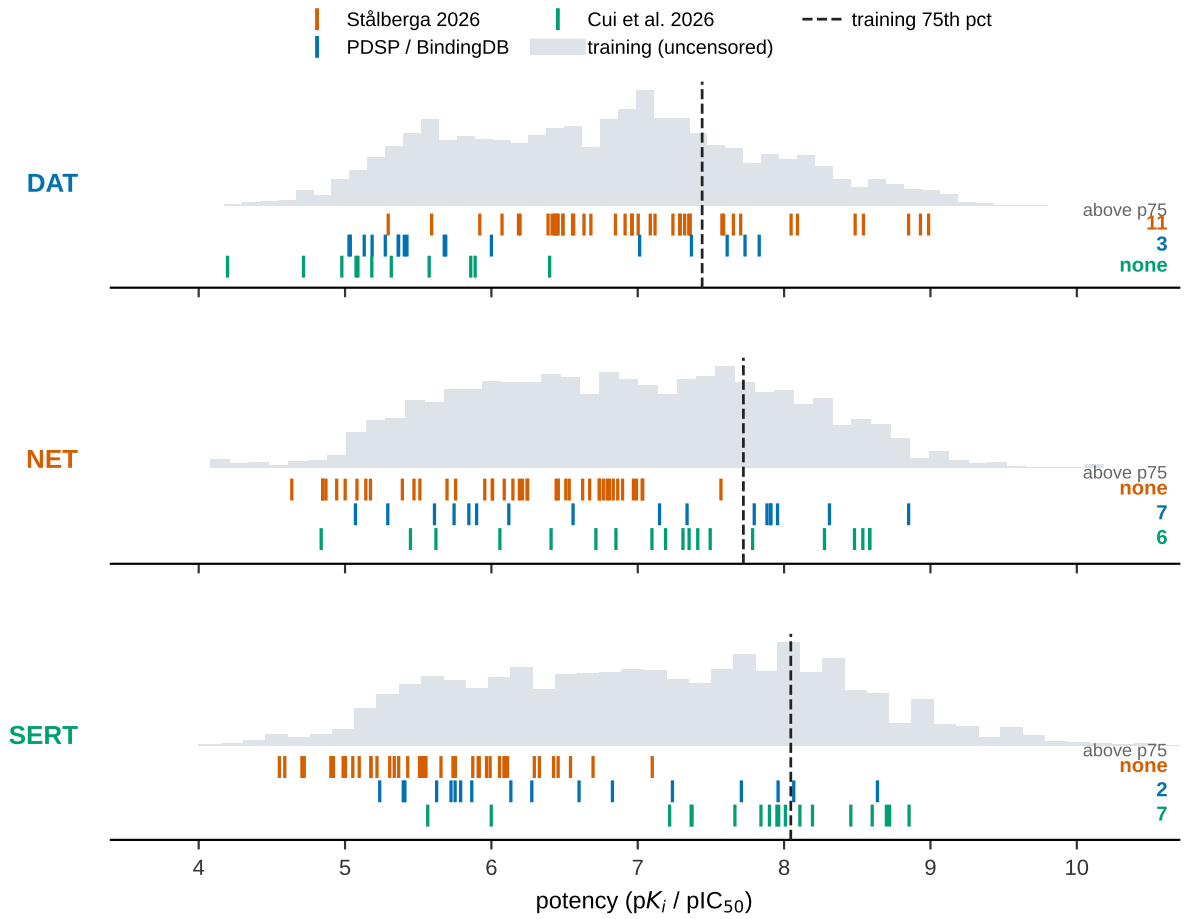


Figure 13: Response-range coverage of the three external sets against the uncensored training distribution, after removing compounds sharing an InChIKey skeleton block with training (42, 17 and 20 compounds remain). Ticks are individual compounds; the dashed line is the training 75th percentile. The advance-specified stimulant panel places no compound above it at NET or SERT, and eleven at DAT. Of the two later sets, PDSP places 7 and Cui 6 above the line at NET, and 2 and 7 at SERT. The advance-specified stimulant panel is the only one of the three confined to a single target.

Spearman $\rho = +0.71$ and $+0.67$, ordering 84% and 83% of compound pairs separated by more than 0.5 log units correctly.

The PDSP set does not reproduce this once its stereochemical duplicates of training compounds are removed. On the remaining 17 compounds it reaches $R^2 = 0.300$ at DAT ($p = 0.023$) but 0.181 at NET ($p = 0.089$) and 0.017 at SERT ($p = 0.62$). Its higher earlier values at those two targets, 0.197 and 0.352, were carried by the thirteen duplicates. One 20-compound series from one laboratory is therefore the whole of the evidence that the NET and SERT shortfall is a property of the test set rather than of the model, and we report it as such. These results do not alter the advance-specified outcome, which stands as reported; they bound its interpretation, and the bound is loose.

Table 15: Achievable R^2 depends on the test set. Ceilings are $1 - \sigma_e^2/\sigma_y^2$, with σ_y^2 the variance of the set being scored and σ_e the dispersion of the measurement relative to the quantity the model targets. *Which σ_e depends on the set's modality.* The two uptake-IC_{50} sets are scored against a binding-dominated training median, so the modality offset applies in full and σ_e is the cross-assay dispersion measured on the independent re-retrieval of § 2.11: SD 0.445, 0.655 and 0.492 at DAT, NET and SERT, over 89, 81 and 71 compounds measured both ways. The PDSP/BindingDB rows are all K_i and therefore match the training modality, so the cross-modality term would be the wrong one; those three rows use the within-binding replicate dispersion instead, 0.452, 0.457 and 0.463. That is the *pooled* estimate — the degrees-of-freedom-weighted root mean of the within-group variances — and not the median of per-group standard deviations reported in Table 1, which is a descriptive statistic and is biased low here because most replicate groups hold two records. On the same pooled estimator the all-modality dispersion is 0.444, 0.471 and 0.525, so restricting to binding lowers it at NET (0.471 → 0.457) and SERT (0.525 → 0.463) and moves it by less than 0.01 at DAT (0.444 → 0.452) — the direction the modality-offset argument of § 2.6 predicts, and largest at the target where that offset is largest. The two agree at DAT and SERT and differ at NET, which is where the modality offset lives; using the cross-assay term uniformly would put the PDSP NET ceiling at 0.698 and its achieved fraction at 26% rather than 0.853 and 21%, i.e. it would flatter the model. These are not the frozen-set dispersions of Table 7 (0.529, 0.658, 0.497), which are computed on different compound sets; the two agree at NET and SERT and differ at DAT, where the frozen value would put the advance-specified set's ceiling at 0.635 rather than 0.741 and the achieved fraction at 72% rather than 62%. *var* is that variance as a fraction of the uncensored training variance. % is the achieved value as a fraction of the ceiling. All values recomputed with the deposited model, with training overlap removed on the InChIKey skeleton block in every set; the advance-specified rows reproduce Table 3 exactly.

Evaluation	Target	n	var	R^2	ceiling	%
Internal, scaffold-disjoint CV						
	DAT	2849	1.00	0.584	0.820	71
	NET	2849	1.00	0.501	0.633	79
	SERT	2849	1.00	0.541	0.838	65
Advance-specified external — Stalberga et al. 2026, forensic stimulants						
	DAT	42	0.70	0.459	0.741	62
	NET	42	0.51	0.015	0.279	5
	SERT	42	0.25	0.058	0.354	16
Exploratory external — PDSP/BindingDB, antidepressant class, binding K_i						
	DAT	17	0.99	0.300	0.812	37
	NET	17	1.21	0.181	0.853	21
	SERT	17	0.76	0.017	0.812	2
Exploratory external — Cui et al. 2026, SNRI series, uptake IC_{50}						
	DAT	11	0.34	0.089	0.466	19
	NET	19	1.06	**0.342**	0.655	**52**
	SERT	18	0.52	**0.468**	0.687	**68**

4 Discussion

One failure, three times. The results above are easier to hold together than their number suggests, because three of them are the same failure. In each case a pooled statistic was read as a property of the thing it names, and turned out to be a property of the composition of the set it was averaged over.

The binding-versus-functional offset at NET reads as a property of the measurement format. It is a property of the assay cell, indexed here by target and radioligand: +0.507 with [^{125}I]RTI-55, +0.038 with [^{3}H]nisoxetine, and spanning 1.5 log units across the binding assays in this corpus (§ 3.8). The per-target number is a weighted average over whichever radioligands dominate that target's literature, and NET looks special because NET is where the largest-offset cell is most used.

The external R^2 reads as a property of the model. It is substantially a property of where the test compounds sit in the training response distribution: restricting the training set to the external response region reproduces 53–90% of the shortfall, and the achievable ceiling on the advance-specified set is 0.279 at NET against 0.633 internally (§ 3.11).

And the composition-matched comparator reads as a property of the model's difficulty. It does not exist: for all nine cells the external set's composition lies outside what any internal subset attains, and the sampler that appears to supply one is widening its tolerance until something is accepted (§ 3.4).

The practical recommendation that follows is one sentence: report what a number was averaged over — which assay cells, which response region — alongside the number, and check that a comparator exists before interpreting a comparison. Everything else here is an instance.

What the radioligand result does and does not say. It does not say the NET offset is an artefact. The effect is large, replicates across 12 independent documents within the RTI-55 stratum, and survives a within-document design that removes between-laboratory calibration. What it says is that "assay format" is the wrong name for it, and therefore the wrong generalisation: a reader who takes a corpus-wide binding-versus-functional correction from this paper and applies it to a NET dataset measured with [^{3}H]nisoxetine would make the prediction worse, not better. That is also, finally, a mechanism for the paper's own negative result that a constant

correction failed to transfer. The quantity is not constant. It is near-constant within an assay cell and varies by more than a log unit between them, which is exactly the condition under which a pooled correction cannot help.

The confound is not resolved here. Radioligand is collinear with document and nearly with chemotype in this corpus, and the experiment that would separate them — the same compounds at NET by both radioligands in one laboratory — has not been done. Our claim is the weaker one that the pooled estimate is an average over heterogeneous cells, which is enough to invalidate the format reading without establishing the radioligand reading.

Scaffold splitting and the sources. Scaffold splitting is conservative with respect to analogue leakage and silent with respect to other sources of external shortfall. Our decomposition indicates that the dominant one is where the test compounds sit in the training response distribution, and that source identity contributes little *unique* variance conditional on the response-range measure included — 2.1%. Because response-range position and source identity are collinear by construction, that is a statement about this regression and not evidence that laboratory effects are unimportant.

Chemical novelty performed even more poorly as an explanation. Chemotype matching reproduced 11–16% of the shortfall, nearest-neighbour similarity contributed 2.2% uniquely, and no external compound lay beyond the training 95th percentile in descriptor space.

How much stringency does a scaffold split buy? Two observations here bear on the wider debate. Measured directly, scaffold splitting cut the proportion of test compounds with a close training neighbour from 74.5% to 26.2% — a real reduction, but one that leaves a quarter of the test set with an analogue in training. And for this target family the Murcko abstraction is a particularly coarse instrument, since it discards the ring substituents that determine DAT/SERT selectivity: positional isomers with materially different profiles reduce to the same scaffold. The remedy is not to abandon scaffold splitting but to report the residual near-neighbour rate alongside it, and to treat similarity-cluster splitting as the stricter comparator.Guo et al. (2024); Fooladi et al. (2025) We note that the model's features are substituent-aware even where the splitter is not, so this concerns the stringency of the evaluation rather than the information available to the model.

Which metric, when. R^2 compares models on a single test set but is not comparable across test sets of differing response spread, because its reference is the deviation of observations from their own mean. RMSE measures absolute accuracy on an interpretable scale and is insensitive to spread, but gives no sense of whether that accuracy is useful for the range at hand. The r_m^2 family was introduced precisely to bridge this gap Roy et al. (2009); Mitra et al. (2010) and remains little used. Our recommendation is to report RMSE and R^2 together with a range-matched internal comparator, so that a reader can see both the absolute error and how much of any R^2 shortfall is attributable to the test set's position.

Report the ceiling, not only the score. The point above can be made quantitative, and we suggest it should be. Measurement heterogeneity caps achievable R^2 at $1 - \sigma_e^2/\sigma_y^2$, and because σ_y^2 is the variance of the evaluation set, that cap moves when the test set does. Here it falls from 0.633 to 0.279 at NET purely by changing which compounds are being predicted (Table 15). A reader given 0.015 and nothing else will conclude the model learned nothing; a reader given 0.015 against a reachable 0.279 learns that the test was close to uninformative before the model was applied, and that the model still underperformed on what remained. Both conclusions are warranted and only the second is actionable. The arithmetic requires one quantity that most benchmarks could supply and few do: an estimate of replicate or cross-modality dispersion on the training endpoint.

The ceiling is not new either. $1 - \sigma_e^2/\sigma_y^2$ is the correction for attenuation due to unreliability, which dates to Spearman and underlies the reliability literature in psychometrics.Spearman (1904) Naming it matters because the name carries the assumption: the classical correction treats measurement error as random, independent of the true value, and mean-zero. The cross-modality dispersion used here is not that. It is the dispersion *about* a systematic shift — the mean offset is excluded, so the shift is not being counted as noise — but the residual dispersion is itself heteroscedastic across assay cells, as § 3.8 shows, and the classical assumption is therefore approximate. Reporting the bound both ways makes the sensitivity visible: with σ_e as the root mean square deviation, which counts the systematic shift as irreducible, every ceiling falls and the NET ceiling on the advance-specified set goes to -0.21, i.e. no model could score above zero. With σ_e as the SD, as in Table 15, it is $+0.279$. The first is too harsh, since a constant shift is removable in principle; the second is the generous bound and is the one we report. The true bound lies between, and nothing in this paper's argument depends on which.

What this means for a forensic or regulatory user. The Introduction names that reader, and the paper owes them a plain answer. One cell of this benchmark has demonstrated external validity: DAT prediction on stimulant-like chemotypes, at $R^2 = 0.459$ against a reachable 0.741, which is 62% of what the measurement noise on that test set allows. Nothing here supports using a NET or SERT prediction for a novel compound with any stated confidence: the advance-specified test could not distinguish either from zero, and the only external set on which both become estimable is a single 20-compound series from one laboratory. A practical reading is that these models can rank a new stimulant's DAT potency against known comparators, which is often the question a scheduling decision turns on, and cannot yet be relied on for the DAT/SERT ratio that separates stimulant from entactogen character, because the SERT half of that ratio is the half with no advance-specified external support, and the one exploratory set that does reach it is the 20-compound series.

A tractable route to NET and SERT coverage. The scarcity of potent external NET and SERT data is structural rather than accidental, and the two sets we assembled afterwards — neither of them large, and only one of them informative at both targets — are the exception that shows how thin the supply is. Assays that yield single-digit nanomolar human SERT or NET potencies are run predominantly by medicinal-chemistry groups publishing in journals that ChEMBL curates comprehensively, so those compounds become training data rather than test data; the pharmacology and forensic-toxicology literature that ChEMBL does not systematically curate concerns stimulants, which are DAT-biased and weak at the other two transporters. That observation also indicates the remedy. ChEMBL_37 was released 2026-05-01 with a literature freeze near July 2025, so any qualifying measurement published after that date is external by construction. A temporal holdout of this kind requires no new experiments, is immune to the exact-match and analogue leakage that scaffold splitting only partly controls, and — unlike the stimulant panels currently available — can be assembled specifically to span the potent end of the NET and SERT range.

Does the model beat its own training data? For the four compounds present in both the training set and an independent study, prediction error against the independent measurement (RMSE 0.878) exceeded the disagreement between the two laboratories measuring the same molecules (0.683), which in turn exceeded the model's error against the values it was trained on

(0.277). At $n = 4$ this is a design rather than a result, but it frames a question that external validation rarely asks: whether a model adds information beyond the training measurement itself. Where inter-laboratory disagreement approaches the apparent generalisation gap, measurement provenance rather than chemical novelty bounds achievable external performance. Assembling a larger set of such compounds is a tractable next study.

A measurable difference that is not an actionable one. The NET binding-versus-functional offset is the clearest positive result here, and also the clearest illustration of a gap between detecting an effect and using it. It survives the controls that hold the laboratory, the mechanism and the value type fixed: within single publications, within uptake blockers alone, across 49 documents, and after separation from the K_i-versus-IC_{50} component with which ChEMBL confounds it. It does not survive stratification by radioligand (§ 3.8), which is the control this paper adds.

And none of the three natural responses recovers any accuracy — correcting by the measured constant worsens external fit, training modality-specific models changes internal accuracy by $+0.002$, and selecting the matching modality at test time costs -0.017 once training-set size is controlled. Two things account for that, and they are different. The first is within-cell: an offset of $+0.26$ sits inside a compound-to-compound spread of 0.658, so even a correct correction is small relative to the scatter it would be applied against. The second is between-cell, and is what the radioligand result adds: the quantity being corrected for is not one quantity. Reviewers of curated bioactivity sets are often asked to correct for such offsets; this is a case where the offset is real, well characterised, and still not worth correcting — and where the reason it cannot be corrected is also the reason it was mis-named.

Two diagnostics that return a number regardless. The outcome-conditioned error regression and the composition-matched comparator fail in the same way and are worth stating together. In the first, the predictor is computed from the observed external value and the outcome is computed from the same value, so a regularised model's shrinkage reproduces the apparent effect with no chemistry involved: $R^2 = 0.444$ against a shrinkage null of 0.448, collapsing to 0.001 in the only form available before the outcome is known. In the second, the comparator is assumed to exist and for none of the nine cells does it, yet the sampler that constructs it returns a band in every case because its tolerance widens until one is accepted. Neither failure

announces itself. In both, the diagnostic produces a plausible number whose magnitude is set by the construction rather than by the data, and in both the check is cheap: compare against a shrinkage null, and confirm that the matched quantity is attainable before matching on it. We obtained three mutually inconsistent headline counts from four runs of the second analysis before running that check, which is the strongest argument we can offer for running it first.

Five recommendations follow. Report absolute error alongside R^2, and consider the r_m^2 metrics designed for this failure mode.Roy et al. (2009) Report a range-matched internal comparator *and the evidence that one exists.* Report the position of the external set within the training response distribution. Disaggregate by source — not because laboratory identity is itself explanatory, but because pooling conceals the range heterogeneity that is. And where assay modality is pooled, label it from the assay description rather than from `assay_type`. That field is specific but insensitive here: it recovers 6% of the functional measurements the assay text identifies. The gap is a definitional mismatch — the field records how the source publication classified the assay — rather than curation error, and it is a reason to read the text, not a criticism of the resource.

5 Conclusion

External validation is widely treated as a single binary test that a model either passes or fails. The present results indicate that the outcome depends heavily on a property of the test set that is rarely reported: where its compounds sit within the training response distribution. A model that appears to fail externally may simply have been evaluated on a narrow slice of its response domain, and one that appears to succeed may have been evaluated on a favourable one. Neither can be distinguished from a single R^2 value.

At NET and SERT, the external data available when this protocol was fixed contained no compound potent enough to evaluate a model at all, and the two sets we found afterwards place the point sharply: one of them restores both targets to significance, and it is a single 20-compound series. Where a target's informative chemotypes are absent from the available external data, external validation is not underpowered but structurally unavailable, and reporting that plainly is more useful than reporting a coefficient indistinguishable from zero. The corollary is that this condition is a property of the literature at a moment in time, not of the target, and

it can lift — which is an argument for stating the coverage alongside the score rather than for treating any one external result as final.

6 Limitations

Five further limitations are declared where they arise rather than here, because each bears on a specific result: the protocol's self-attested status and its two post hoc amendments (§ 2.8); the measured accuracy and the error structure of the modality classifier (§ 2.2); the overlap filter's change and what it costs the PDSP set (§ 3.12); the collinearity of radioligand with document and chemotype, which is why the stratification identifies a composition dependence and not a mechanism (§ 3.8); and the unattainability of a composition-matched comparator in every cell (§ 3.4). What follows is everything else.

What the external evaluation can support.

- The sample could not distinguish null from modest external association at NET and SERT.
- The extended evaluation is exploratory; its endpoints were not pre-specified.
- Two of the four extended sources — the Luethi and Liechti compilation and the Basel primary studies — share a laboratory of origin.
- The extended external set spans 114 compounds but only 31 Murcko scaffolds (12, 11, 14 and 3 per source), so the decomposition is established over a narrow slice of chemical space and its generalisation to more diverse external sets is untested.
- The nominal spread in R^2 across sources did not reach significance ($p = 0.065$). This is a borderline result on four sources and should not be read as demonstrating the absence of a laboratory effect; per-source estimates are reported for completeness, not as evidence of one.
- The source-effect decomposition rests on four studies; the 2.1% figure indicates that source adds little beyond range rather than being a precise estimate.
- The familiar-chemistry comparison has $n = 4$ and is a design rather than a result.
- Conditioning the model on assay modality was tested on the labelled corpus rather than on the advance-specified benchmark, so that null result is exploratory and shares the external

sets' limited coverage at NET and SERT.

The measurement and the curation.

- Censored values are carried at their bounds; a full Tobit treatment was not adopted.
- Seven external compounds were lost to non-determination entries without numeric bounds.
- 15.8% of compounds lie closer to a selectivity class boundary than one replicate measurement resolves.
- Murcko scaffolds are a coarse grouping for this target family, since the abstraction discards the ring substituents that determine selectivity; the scaffold-disjoint folds are correspondingly a weaker stringency guarantee than the scaffold count alone suggests.
- Chemotype matching was by compound count rather than by matching the full similarity distribution.

What the offset analysis does not settle.

- The offset varies roughly threefold across chemotypes (Kruskal–Wallis across strata, $p < 0.001$). The pooled value is an average over a heterogeneous set, and the per-stratum samples ($n = 5$ to $n = 25$) are too small to support a chemotype-resolved correction.
- The chemotype stratification was performed on the independently re-retrieved set, which is non-nested with the frozen one. It therefore tests whether the effect is chemotype-specific in general, rather than partitioning the frozen estimate.
- The DAT-versus-SERT comparison is sensitive to retrieval and curation: the two are not distinguishable in the frozen set ($p = 0.39$) but are on an independent re-retrieval ($p = 0.005$). Only the contrast between NET and the other two targets is claimed as robust.
- The fixed-effects decomposition broadens the document base from five papers to 49, but it still cannot separate assay modality from anything that co-varies with it inside a single report — notably substrate concentration and incubation time, which are fixed per study.
- The measured offset is established among uptake blockers. The corpus is overwhelmingly blockers, the structural classifier of § 2.3 assigns only 42 compounds as substrate-type, and

the offset should not be assumed to transfer to substrate-type compounds, where the same ratio is one to two orders of magnitude larger.

- The models predict potency only and do not distinguish transporter substrates from pure uptake blockers, which is the primary pharmacological distinction for this target family and is not recoverable from a potency value alone. A criterion based on the ratio of binding affinity to uptake potency exists Eshleman et al. (2017) but was outside the advance-specified endpoints, and using it to stratify the offset analysis would be circular, since that ratio is the quantity under test. Models of these targets that do not separate the two are predicting a composite endpoint.

Inference.

- No multiplicity adjustment is applied. The NET-versus-other-targets contrast is the one confirmatory comparison; every other p-value in this paper is exploratory and should be read as such. The confirmatory contrast survives any correction ($p = 1.2 \times 10^{-8}$ on the re-retrieval, 2.7×10^{-11} on the frozen set); several exploratory results rest on p between 0.005 and 0.05 and would not.
- The external-set ceilings in Table 15 inherit every assumption behind the dispersion estimates they use: equal error variance in the two modalities, independent errors, and homoscedasticity across the response range. They are order-of-magnitude bounds on what a test set of that composition could have shown, not precise values.
- Response-range position and source identity are collinear by construction; the partition separates them statistically but not causally.

Supporting Information

A companion analysis of structure-based prediction at these targets — redocking across nine transporter–ligand complexes, two scoring functions,Trott and Olson (2010); Eberhardt et al. (2021); Quiroga and Villarreal (2016) cross-docking between conformational states, and pose-derived interaction features — reaches conclusions consistent with an independent report using different software.Pazos et al. (2026)

List of abbreviations

DAT: dopamine transporter; NET: norepinephrine transporter; SERT: serotonin transporter; CCC: concordance correlation coefficient; CI: confidence interval; ECFP: extended-connectivity fingerprint; IQR: interquartile range; MAE: mean absolute error; RMSE: root-mean-square error; SD: standard deviation; SEM: standard error of the mean.

Declarations

Availability of data and materials

The datasets supporting the conclusions of this article are available in the Zenodo repository, 10.5281/zenodo.23144306. The repository contains the curated benchmark (2849 compounds with per-target potencies, censoring flags, scaffolds and provenance), the frozen fold assignments, the extended external set with source attribution, and all analysis code, released under CC BY 4.0 for data and the MIT licence for code. The revision additionally deposits the independent re-retrieval script (`src/replicate_assay_offset.py`), the 16 750 retrieved activity records, the per-compound paired NET table, and the chemotype stratification output in machine-readable form. No registration or licence agreement is required for access. Bioactivities derive from ChEMBL release 37 (and, for the re-retrieval, from the live release accessed 2026-09-27); structures from PubChem.

Competing interests

The author declares no competing interests.

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Authors' contributions

SES conceived the study, curated the data, performed all analyses, produced the figures, and wrote the manuscript. The author read and approved the final manuscript.

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Implementation errors identified during development

In the interest of reproducibility we record errors found and corrected before submission: a protonation routine that left tertiary amines neutral; a use-after-free in structure preparation producing non-deterministic coordinate corruption; a misattributed prior-art citation; a reproducibility statistic inflated four-fold by taking a maximum of ranges rather than a per-target standard deviation; a selectivity threshold attributed to the wrong publication; a fold-generation function that silently ignored its random seed; twenty CAS registry numbers leaked across wrapped table rows during structure resolution; and a database property rename that caused a structure cross-validation check to return zero comparisons. Each produced plausible output and raised no error, and all were concentrated in variance estimation, provenance handling or validation code rather than in point estimates. Automated regression tests covering each are included in the deposited repository.

A further set was found during revision and each is declared where it bears on a result rather than only here: one external structure recorded as the wrong molecule, with its two linked consequences (§ 2.8 and Limitations); an overlap filter applied on the full InChIKey where the fingerprint ignores stereochemistry, which left 13 compounds of one exploratory set fingerprint-identical to training compounds (§ 3.12); a figure that plotted the reciprocal of the regression whose slope it annotated (Figure 6); two figure annotations asserting claims the body retracts; a reproducibility statement claiming three-decimal agreement across values differing at the third decimal (§ 2.7); a structure file in which two records carried a CXSMILES annotation the default parser rejects, silently dropping one real compound; and several arithmetic slips in counts and standard-deviation ratios. The pattern is the same as above — each produced plausible output

and none raised an error — and the two that changed a reported result, the overlap filter and the radioligand stratification of § 3.8, both moved it against the paper's prior claim.

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